

Prefix Instructions and Incompressible Flows

OpenAI

September 27, 2026

Abstract

Finite prefix instructions may change area and erase information. We give explicit history processors and smooth incompressible motions that retain that information and realize every instruction on its full domain. A first application assigns each machine and finite input a smooth mean-zero force on the flat unit three-torus, at any fixed positive computable viscosity. The solution starts from rest; a fixed particle enters a fixed open strip exactly when the machine halts. The force repeats with period one after an initial loading interval. We then prove alternative realizations using full boxes, normal compensation, invariant planes, and a spatial clock. The constructions specify their initialization, comparison class, derivative bounds, and continuous-time observation. Exact formulas and effective cutoffs provide finite descriptions of the fields and all their derivatives.

1 Introduction

A finite machine instruction reads a short part of a configuration and changes it. A material flow, by contrast, must be invertible at every finite time. Two questions therefore arise when a particle is to carry a computation: where is the information erased by an instruction retained, and how is the resulting injective update made compatible with incompressibility? We study these questions for finite prefix instructions and their smooth three-dimensional realizations.

A stack is an infinite word whose first letter is its accessible end. A prefix instruction has the form

$$(s; v, w) \mapsto (s'; v', w'),$$

meaning that it sends (s, vL, wR) to $(s', v'L, w'R)$ for every pair of unchanged tails L, R . Here s, s' belong to a finite set of control states, and all four prefixes are finite words over a finite alphabet. A positional code turns this operation into a positive diagonal affine map between two rectangles. The source rectangles must be separately disjoint, and so must the target rectangles. Intersections between a source rectangle and a target rectangle are allowed. These conditions are stronger than injectivity along one initialized computation.

The planar map need not preserve area. Appending one history letter, for example, contracts one stack coordinate without expanding the other. We realize such instructions in two ways. A compensating normal strain transports the entire coded rectangle and a three-dimensional neighborhood. Alternatively, a compressible planar motion is placed in an invariant plane of an incompressible field. The latter preserves the prescribed planar dynamics without prescribing an affine map off the plane. Both methods require control of the complete continuous path: correctness at integer sampling times alone does not establish a fixed particle observation.

1.1 The fluid problem and a first result

Let $\mathbb{T}^3 = (\mathbb{R}/\mathbb{Z})^3$ with its flat metric, and fix a positive computable viscosity ν . For an arbitrary fixed positive real ν , the same formulas and conclusions hold, and numerical evaluation is relative to that coefficient. We use

$$u_t + (u \cdot \nabla)u - \nu \Delta u + \nabla p = f, \quad \operatorname{div} u = 0, \quad u(0) = 0. \quad (1)$$

An instance is a deterministic machine with finite state set, finite work alphabet with a blank, an initial state, a specified halt set, and a finite input written on an otherwise blank two-sided tape. Its head starts at cell zero. Missing transitions may be interpreted as halting. For a fixed initial label a , the material position satisfies $X'(t) = u(t, X(t))$ and $X(0) = a$. The observation asks whether $X(t) \in O$ for some $t \geq 0$, where O is a fixed open set.

A finite effective force prescription is a finite program that evaluates f and any requested mixed derivative at computably specified arguments to any positive rational error, and supplies the derivative bounds used on compact sets. The program describes every instruction branch. It does not generate a machine execution and replay that execution as a force. On a torus our classical comparison class requires u, u_t , spatial derivatives of u through order two, and $p, \nabla p$ to be continuous on every finite closed time cylinder. Velocity and pressure are periodic, and pressure has spatial mean zero. The prescribed solutions themselves are smooth.

Theorem 1.1 (Interspersed history on moving sheets). *At fixed positive computable viscosity, every machine and finite input effectively determine a smooth mean-zero general force on the unit $\mathbb{T}^3$, with all mixed derivatives bounded and period one for $t \geq 1$. Its zero-data solution is globally smooth and unique in the classical periodic comparison class, with uniformly bounded kinetic energy. Its material trajectory from $(1/8, 1/4, 1/4)$ enters $\{x : 1/2 < x_1 < 1\}$ at some finite time exactly when the machine halts. The repeated template depends only on the table, and the input enters only the loading pulse.*

This first result is proved with history records interspersed among work letters. Its geometric map changes planar area, so a third scale restores volume. Later constructions change the history arrangement or initialization when a different property is wanted: compact Euclidean support, a bounded detector, periodicity from the initial time, or a spatial clock. The comparison classes, initial labels, detectors and temporal conclusions are stated separately for these alternatives.

1.2 Antecedents and the role of the present constructions

The logical obstruction comes from Turing's undecidable symbol-printing problem [10, Section 8]. Modify such a machine to halt at the first printing of the designated symbol, and redirect any earlier ordinary stop into an infinite dummy loop. It then halts exactly when the designated symbol would have been printed, giving the halting formulation used here. Retaining information lost by an irreversible instruction was discussed by Landauer [5, Section 3] and implemented by Bennett's history recording [1, Eqs. (9)–(11)]. Our histories remain stored. We prove the local tables and their inverses explicitly; the cleanup and complexity statements of Bennett's reversible simulation are not used.

Moore's generalized shifts connect finite symbolic instructions with positional codes and piecewise affine dynamics [7, 8]. The smooth realization and suspension for invertible generalized shifts in [8, Section 6, Theorem 12 and its corollary] are direct antecedents of the present viewpoint. We need additional statements on full closed rectangles, effective derivative bounds, incompressibility, zero initial velocity, and observation at every intermediate real time. These are established by the

explicit motions below. The classical fragility of positional tape storage also explains why an exact event should not be read as a uniform finite-precision tolerance.

Cardona, Miranda, Peralta-Salas and Presas constructed Turing-complete stationary Euler particle flows on a three-sphere with a chosen metric [3]. Their area-preserving extension of generalized shifts uses contraction, transport and expansion of separated blocks followed by a relative area correction [3, Proposition 5.1]. That construction is an important predecessor of the planar routing used here. Our explicit normal compensation and invariant-plane formulas give the prescribed flat three-dimensional motions directly; their full neighborhood and detector properties are proved at the point of use. Cardona, Miranda and Peralta-Salas also constructed Turing-complete stationary Euler flows for the standard Euclidean metric on $\mathbb{R}^3$, with computation on a noncompact invariant plane and infinite kinetic energy [2, Theorem 1 and the discussion following it].

Universality is also known with viscosity. Dyhr, González-Prieto, Miranda and Peralta-Salas construct unforced stationary Navier–Stokes particle universality on suitable compact three-manifolds after deforming the metric, using Hodge viscosity and a harmonic velocity [4, Theorem A]. The fixed flat domains, prescribed external force, and zero initial velocity in the present paper specify a different fluid experiment. Our results concern these constructed solutions and exact material events; they make no assertion about arbitrary-data regularity.

The companion *Finite Instructions and Solenoidal Shear Flows* [9] gives a complete initialized construction with reciprocal planar maps and directly solenoidal force at pressure zero. We use its full-domain recorder in one later specialization, with an explicit identification of all guards. The general residual forces below need not be solenoidal. When a torus force is projected, the accompanying pressure changes; that operation is kept distinct from the direct shear identity.

1.3 Proof strategy and reading order

The first application has four steps. Gapped radix intervals turn complete prefix cylinders into separated closed rectangles. A local curl formula then implements an affine volume-preserving motion on a neighborhood. Private heights separate the instruction rectangles while a reciprocal normal strain corrects their planar area changes. Finally, a history invariant identifies the computation, a loading segment places the fixed particle at its initial code, and the horizontal interpolation proves the detector equivalence throughout every instruction period.

Section 2 records the residual realization and the exact analytic comparison classes. Section 3 develops the radix and moving-sheet tools. Section 4 completes Theorem 1.1. Sections 5 and 6 then explain the additional requirements of full boxes and invariant-plane routing. Sections 7–9 organize the resulting processors by where they retain history: next to a separator, on a tape track, or in a separate stack. Sections 10–12 treat further history placements, boot instructions and fixed observers. Section 13 gives two compact reciprocal realizations. The final section replaces a temporal program by a spatial clock and verifies its separate decay and phase-window observation.

2 Analytic classes and changes of physical coordinates

The geometric constructions prescribe a smooth velocity first. This section records exactly which other solutions are excluded by the energy argument. Pressure assumptions are kept separate: continuity in a pressure norm, a norm at each individual time, and an integrable pressure gradient are different conditions.

Write $\mathcal{F}_\nu[U] = U_t + (U \cdot \nabla)U - \nu\Delta U$. All noncompact results below use $\mathbb{R}^3$; their velocities and forces have one fixed compact spatial support unless the individual statement says otherwise.

This property concerns the constructed solution, not a competing solution. On $[0, T]$, write $CH^k = C([0, T]; H^k(\mathbb{R}^3))$ and $C^1 L^2 = C^1([0, T]; L^2(\mathbb{R}^3))$. Spatial constants in pressure may depend on time.

Lemma 2.1 (The pressure term without a pressure norm). *If $w, g \in L^2(\mathbb{R}^3; \mathbb{R}^3)$ satisfy $\operatorname{div} w = 0$ and $\operatorname{curl} g = 0$ in distributions, then $\langle w, g \rangle_{L^2} = 0$. In particular, any distributional gradient $g = \nabla p \in L^2$ has zero pairing with w , without an assumption that $p \in L^2$.*

Proof. An L^2 field is a tempered distribution. The differential identities, initially tested on compactly supported smooth functions, extend to Schwartz tests by multiplying by cutoffs tending to one and using L^2 convergence of the tests and their first derivatives. Fourier transformation gives $\xi \cdot \widehat{w} = 0$ and $\xi_j \widehat{g}_k - \xi_k \widehat{g}_j = 0$. These are identities of locally integrable functions, so they hold almost everywhere. Away from $\xi = 0$, $\widehat{g}$ is parallel to ξ and $\widehat{w}$ is perpendicular to it. Their Hermitian product is zero. The exceptional point has measure zero; Plancherel proves the assertion. $\square$

Lemma 2.2 (Residual realization and separate comparison classes). *Let $\nu > 0$ and let U be a prescribed smooth divergence-free velocity on $[0, \infty) \times \mathbb{R}^3$, with $U(0) = 0$. On each finite interval, assume $U \in CH^2 \cap C^1 L^2$ and $U, \nabla U$ are bounded. Put $f = \mathcal{F}_\nu[U]$. Then $(U, 0)$ is a solution. Its velocity is unique in each of the following separately stated comparison classes, with the same force and initial data:*

- (i) *Smooth classical $v \in CH^2 \cap C^1 L^2$, with bounded v , and a pressure representative $p \in CH^1$.*
- (ii) *Smooth classical $v \in CH^2 \cap C^1 L^2$, with bounded $v, \nabla v$, and an H^1 pressure representative at each time, with no required time continuity in that pressure norm.*
- (iii) *Strong solutions with $v \in CH^4 \cap C^1 H^2$ and distributional $\nabla p \in CL^2$.*
- (iv) *Smooth classical $v \in CH^2 \cap C^1 L^2$, with bounded $v, \nabla v$, and a pressure representative $p \in CL^2$.*
- (v) *Smooth classical $v \in CH^2 \cap C^1 L^2$, with bounded $v, \nabla v$, without an additional integrability or normalization condition on pressure.*

The pressure gradient is unique. An L^2 pressure representative, when specified, is zero; otherwise pressure is determined up to a function of time. The reference material flow is a smooth diffeomorphism at every finite time and exists for every finite forward time.

On a flat torus $\mathbb{T}_L^3$, the same conclusion holds in the classical class where v, v_t , spatial derivatives through order two, and $p, \nabla p$ are continuous on finite closed cylinders, with periodic mean-zero pressure. No Sobolev conditions at infinity are involved in that assertion.

Proof. Substitution proves existence of the prescribed solution. Set $w = v - U$. The difference equation is

$$w_t + (v \cdot \nabla)w + (w \cdot \nabla)U = \nu \Delta w - \nabla p, \quad \operatorname{div} w = 0.$$

The stated velocity regularity implies $w \in C^1 L^2 \cap CH^2$, so $\frac{d}{dt} \|w\|_2^2 = 2\langle w, w_t \rangle$ at each time. In case (iii), the stronger assumptions imply these inclusions. They also imply bounded velocity and spatial derivatives through order two: for $j \leq 2$, Cauchy–Schwarz in Fourier space uses the finite integral $\int_{\mathbb{R}^3} |\xi|^{2j} (1 + |\xi|^2)^{-4} d\xi$.

For the transport term, insert a cutoff χ_R that equals one on the ball of radius R , is supported in the ball of radius $2R$, and satisfies $|\nabla \chi_R| \leq C/R$. Its boundary error is at most $CR^{-1} \|v\|_\infty \|w\|_2^2$, which tends to zero. Diffusion is integrated by parts in H^2 ; alternatively its cutoff error is bounded by $CR^{-1} \|w\|_2 \|\nabla w\|_2$. The deformation term is bounded by $\|\nabla U\|_{\infty, \text{op}} \|w\|_2^2$.

The same pressure cancellation applies in all five classes. At each time, the equation gives the following equality of distributional gradients in L^2 :

$$\nabla p = f - v_t - (v \cdot \nabla)v + \nu \Delta v \in L^2, \quad \|(v \cdot \nabla)v\|_2 \leq \|v\|_\infty \|\nabla v\|_2.$$

The velocity assumptions in every class give all terms on the right in L^2 ; the identical estimate for U and its stated time and space regularity give $f \in L^2$. Since a distributional gradient is curl free, Lemma 2.1 gives $\langle \nabla p, w \rangle = 0$. This cancellation neither chooses an L^2 pressure representative nor uses continuity of pressure in a norm.

In every case the resulting pointwise energy inequality is

$$\frac{1}{2} \frac{d}{dt} \|w\|_2^2 + \nu \|\nabla w\|_2^2 \leq \|\nabla U\|_{\infty, \text{op}} \|w\|_2^2.$$

The coefficient is bounded on each finite interval and $w(0) = 0$, so an integrating factor gives $w = 0$. The pressure has already disappeared before time integration; the proof imposes no unlisted temporal pressure norm. Substitution then gives $\nabla p = 0$. A spatially constant L^2 function on $\mathbb{R}^3$ is zero, which proves the normalization assertion.

On the torus every integration by parts is periodic and has no boundary term. The specified classical regularity justifies norm differentiation, diffusion, and pressure cancellation. The same energy inequality gives uniqueness and the mean-zero normalization fixes pressure. In either domain, the smooth reference velocity and its bounded spatial derivative give unique material trajectories; bounded speed prevents finite-time escape. Solving backwards on a finite interval gives the inverse flow, and smooth ODE dependence gives its smoothness. $\square$

Cases (i)–(v) identify the precise hypotheses used by the applications; their presence in a common lemma does not replace a theorem's declared comparison class by a different one. In particular, a pressure norm is not silently imposed in a gradient-only application. The proof is the classical energy comparison for prescribed smooth solutions [6, Section 18], with its noncompact pressure pairings made explicit.

For the flow Φ_t , differentiation of its trajectory equation gives

$$U(t) = (\partial_t \Phi_t) \circ \Phi_t^{-1}, \quad \partial_{tt} \Phi_t(a) = (f - \nabla p + \nu \Delta U)(t, \Phi_t(a)).$$

These identities describe the material form of the same constructed solution; they do not add an independent existence claim.

Lemma 2.3 (Physical scaling). *For $a, b > 0$, $x = x_0 + ay$, and $s = bt$, let $U(t, x) = abV(bt, (x - x_0)/a)$. Then trajectories are transformed by this change of variables, incompressibility is preserved, and at the prescribed physical viscosity ν ,*

$$\mathcal{F}_\nu[U](t, x) = ab^2 \left[V_s + (V \cdot \nabla_y)V - \frac{\nu}{a^2 b} \Delta_y V \right] (bt, (x - x_0)/a).$$

Proof. The trajectory equation gives $\dot{x} = abV$; differentiating in space gives $\text{div}_x U = b \text{div}_y V$. The three residual terms scale by ab^2 , ab^2 , and b/a , respectively, proving the formula. $\square$

Thus a side-10 torus is kept as a physical side-10 torus. If a chart is rescaled, the residual is recomputed with the fixed ν ; the equation is not identified with a differently normalized viscosity.

Proposition 2.4 (Projection on a flat torus). *An effective smooth mean-zero force g on $\mathbb{T}_L^3$ has an effective mean-zero potential ϕ satisfying $\Delta\phi = \operatorname{div} g$. Replacing (g, p) by $(g - \nabla\phi, p - \phi)$ preserves velocity and makes the force solenoidal. Uniform mixed-derivative bounds, temporal L^2 bounds of their spatial suprema, and separately imposed time periodicity, eventual stationarity, or orderwise decay bounds are preserved.*

Proof. In frequencies $\xi_k = 2\pi k/L$, set $\hat{\phi}(k) = -i\xi_k \cdot \hat{g}(k)/|\xi_k|^2$ for $k \neq 0$ and zero for $k = 0$. The multiplier from g to $\nabla\phi$ is $\xi_k \xi_k^\top / |\xi_k|^2$. For a spatial derivative order r , integrate each Fourier coefficient $r + 5$ times in a coordinate with largest $|k_i|$. There are $O(n^2)$ lattice points on the shell $\|k\|_\infty = n$, so the derivative series and its tail converge absolutely, with

$$\|\partial_t^h \partial_x^\alpha \nabla \phi\|_\infty \leq C_{\alpha,L} \max_i \|\partial_t^h \partial_{x_i}^{|\alpha|+5} g\|_\infty.$$

This bound is pointwise in time, and proves each asserted norm or weighted time estimate. Derivative bounds give effective Fourier tails; coefficient integrals can be computed by Riemann sums with derivative error estimates. The formula gives the Poisson identity, divergence cancellation, and the pressure sign by substitution. Linearity preserves the time symmetries. $\square$

Projection generally changes pressure. It supplies no compact-support conclusion on Euclidean space and is distinct from the direct zero-pressure solenoidal shear construction.

Effective flat profiles. For later cutoff formulas put

$$\rho(s) = \begin{cases} e^{-1/s}, & s > 0, \\ 0, & s \leq 0, \end{cases} \quad \sigma(s) = \frac{\rho(s)}{\rho(s) + \rho(1-s)}. \quad (2)$$

Positive-side derivatives of ρ are polynomials in $1/s$ times $e^{-1/s}$ and extend flatly at zero. The estimate $v^m e^{-v} \leq (m+k)! v^{-k}$ gives effective error bounds near the seam, and $\rho(s) + \rho(1-s) \geq e^{-2}$. Products, translations, rescalings, and derivatives therefore yield effective cutoffs with all requested derivative bounds. Periodization uses zero collars; at approximate real arguments a finite superset of possible translates is evaluated. No real equality test at a seam or execution of the encoded computation is used.

3 Prefix maps with changing area

A stack operation changes a finite prefix and leaves the remaining word unchanged. In a positional code this becomes a diagonal affine map of a rectangle. The two scale factors need not be reciprocal. We shall therefore use either a normal strain to compensate the area change, or a planar motion embedded in an invariant plane. These two methods have different geometric guarantees: the former moves a neighborhood in three dimensions, whereas the latter preserves one plane throughout the motion.

Throughout these constructions, viscosity is a fixed positive computable number. At an arbitrary fixed positive real viscosity, the same formulas are effective relative to that coefficient: the residual has form $G + \nu J$ with effective coefficient fields independent of ν . Coordinate inequalities on the unit torus denote the displayed arcs modulo one; the path estimates use the stated representatives inside a chart.

3.1 From prefix cylinders to closed rectangles

We use the machine convention of a two-sided tape, a finite alphabet with blank, and moves $-1, 0, 1$. A missing instruction interpreted as halting can be replaced by an instruction that preserves the scanned symbol, stays in place, and enters a new halt state. Multiple halt states can be merged when a construction calls for a single one. A left-end convention is represented by a permanent tag at the initial cell, with writes preserving that tag and the prescribed endpoint rule used there. These finite changes preserve the halting question.

A two-stack configuration is (s, L, R) , with both words written top first. A rule $(s; v, w) \rightarrow (s'; v', w')$ means

$$(s, vL_*, wR_*) \mapsto (s', v'L_*, w'R_*)$$

for arbitrary infinite tails. A *cylinder* is the set specified by the state and the two finite prefixes. We shall always check the full source and image cylinders separately. Agreement on initialized runs alone would not justify a smooth invertible extension.

Lemma 3.1 (Gapped stack coordinates). *Choose an integer $B \geq 2$ and a finite digit set in $\{0, \dots, B-2\}$ whose distinct elements differ by at least two. For words over the corresponding alphabet define*

$$E(v) = \sum_{j=1}^{|v|} d(v_j)B^{-j}, \quad I(v) = [E(v), E(v) + B^{-|v|}], \quad E(L) = \sum_{j \geq 1} d(L_j)B^{-j}.$$

The infinite code is injective and satisfies $E(vL) = E(v) + B^{-|v|}E(L)$. Incompatible finite prefixes give positively separated closed intervals. If separately disjoint state rectangles are parameterized by positive diagonal affine maps P_s from $[0, 1]^2$, then separately disjoint source and image cylinders give separately disjoint closed source and target rectangles

$$P_s(I(v) \times I(w)), \quad P_{s'}(I(v') \times I(w')).$$

The positive diagonal affine map between a matched pair implements the rule on every encoded configuration in its source cylinder.

Proof. Prefixing a symbol selects the interval $(d + [0, 1])/B$. These intervals lie in $[0, 1]$, and two of them have a gap at least B^{-1} . Further prefixing selects nested affine copies. At the first difference in position j , the gap is at least B^{-j} , which proves both interval separation and injectivity. The series identity follows by splitting its first $|v|$ terms. Two product cylinders with the same state are disjoint precisely when the prefixes on at least one stack are incompatible: otherwise their compatible prefixes have common infinite extensions. This proves rectangle separation. Finally the affine map preserves each normalized tail coordinate $E(L_*), E(R_*)$, which proves its action on codes. Codes of $v\Box^\infty$ are rational, equal to $E(v) + B^{-|v|}d(\Box)/(B-1)$. Digit gaps allow recovery of every finite prefix to sufficiently high finite precision, including zero-digit boundary codes. $\square$

The history mechanism follows Bennett's information-retaining approach [1]; the use of positional codes is closely related to Moore's generalized shifts [7, 8]. The additional work here is the full-cylinder inverse check and a smooth incompressible extension with controlled intermediate trajectories.

3.2 Local motion and normal compensation

Write σ for the effective flat step of (2) and $\theta(s) = \sigma(3s - 1)$. All schedules below are finite rescalings of these functions. For an interval $[a, b]$ and margin $\rho > 0$, the product

$$\sigma\left(\frac{x - a + \rho}{\rho/2}\right) \sigma\left(\frac{b + \rho - x}{\rho/2}\right)$$

equals one on its $\rho/2$ enlargement and is supported in its ρ enlargement. Products in the coordinates give box cutoffs. The effective flatness estimates in Section 2 justify evaluation of all mixed derivatives across their seams; no equality test at a seam is used.

Lemma 3.2 (Localized affine motions). *Let $c(t) \in \mathbb{R}^3$ and a positive diagonal matrix $D(t)$ be smooth effective functions on a compact interval, with $D(t_0) = I$ and $\det D(t) = 1$. Put $A = \dot{D}D^{-1}$. Let K_0 be a rectangular box, allowing zero thickness in a coordinate. If an effective smooth compactly supported cutoff χ equals one on a positive collar of the moving box $K_t = c(t) + D(t)(K_0 - c(t_0))$, then*

$$V(t, x) = \nabla \times \left\{ \chi(t, x) \left[\frac{1}{2} \dot{c} \times (x - c) + \frac{1}{3} (A(x - c)) \times (x - c) \right] \right\} \quad (3)$$

is divergence free and its trajectory from $x_0 \in K_0$ is

$$x(t) = c(t) + D(t)(x_0 - c(t_0)). \quad (4)$$

The formula holds on an effective positive neighborhood of K_0 . Disjoint moving supports allow finitely many such motions to be summed.

Proof. Differentiating $\det D = 1$ gives $\text{tr } A = 0$. With $Y = x - c$,

$$\nabla \times (\dot{c} \times Y) = 2\dot{c}, \quad \nabla \times ((AY) \times Y) = 3AY.$$

The second identity follows by expanding the curl and using $\text{div}(AY) = 0$. Thus on the plateau $V = \dot{c} + AY$, and (4) solves its ODE. If $M \geq 1$ bounds $\|D\|$ and the collar width is η , an initial perturbation smaller than $\eta/(2M)$ remains on that plateau. For a finite sequence, decrease this radius by the product of the preceding affine norms. Smooth ODE uniqueness proves the assertion, and disjoint supports prevent interference. $\square$

Lemma 3.3 (Transport of area-changing sheets). *Let B_i and C_i , $1 \leq i \leq N$, be two separately positively separated finite families of nondegenerate rational closed rectangles in $(0, 1)^2$. Let $F_i : B_i \rightarrow C_i$ be the positive diagonal affine map, and let $z_0 \in (0, 1)$ be rational. There is an effective smooth divergence-free field supported in $(0, 1) \times (0, 1)^3$ whose time-one map agrees on a positive three-dimensional neighborhood of each $B_i \times \{z_0\}$ with a diagonal affine map restricting to the map $(x, z_0) \mapsto (F_i(x), z_0)$, with normal scale reciprocal to its planar area multiplier. On the sheet, each horizontal coordinate is constant during ascent and descent and is a convex combination of its two endpoint coordinates during the middle stage. The same construction applies in a larger bounded Euclidean region after translations and a choice of separated heights.*

Proof. For $N = 0$ take zero. Write p_i, q_i for the centers and w_i^0, w_i^1 for side-length vectors. In three successive stages, lift to $z_i = 1/2 + i/[4(N + 1)]$, transform there, and lower to z_0 . During the middle stage, with a flat progress parameter $s \in [0, 1]$, put

$$p_i(s) = (1 - s)p_i + sq_i, \quad w_i(s) = (1 - s)w_i^0 + sw_i^1,$$

$$D_i(s) = \text{diag} \left(\frac{w_{i1}(s)}{w_{i1}^0}, \frac{w_{i2}(s)}{w_{i2}^0}, \frac{w_{i1}^0 w_{i2}^0}{w_{i1}(s) w_{i2}(s)} \right).$$

All scales are positive and their product is one. Give the sheets a sufficiently small rational thickness. The reciprocal scale is bounded by $\max_i [\min(1, w_{i1}^1/w_{i1}^0) \min(1, w_{i2}^1/w_{i2}^0)]^{-1}$, so an effective positive thickness keeps the middle-stage boxes apart vertically. During lifting use the source separation, and during lowering the target separation. Choose a rational collar smaller than a quarter of the corresponding coordinate gaps and boundary clearances. Lemma 3.2 now supplies the three local motions on disjoint supports, including a positive initial neighborhood. A sheet point has zero normal displacement relative to its center. Its fixed fractional coordinate in the interpolated interval gives $(1-s)x_a + sF_i(x)_a$ in each horizontal direction. Flat stage collars finish the construction. The larger-region version uses exactly the same finite clearance argument. $\square$

3.3 Closing the fluid argument

Lemma 3.4 (Realization and comparison). *Let U be an effective smooth divergence-free field, $U(0) = 0$, on the unit torus or with fixed compact spatial support on $\mathbb{R}^3$. Assume its mixed derivatives are bounded on each finite time interval. Then*

$$f = \partial_t U + (U \cdot \nabla)U - \nu \Delta U \tag{5}$$

is an effective prescribed force and $(u, p) = (U, 0)$ is a global solution. Its velocity is unique in the applicable comparison classes of Lemma 2.2; pressure is determined with exactly the normalization or additive time function specified there. Uniform mixed-derivative bounds, a time period, fixed compact support, and spatial mean zero pass from U to f when imposed on U . The material flow is defined for every finite time.

Proof. The compactly supported field satisfies the whole-space velocity class $C_t H^2 \cap C_t^1 L^2$ and has bounded velocity and gradient; zero pressure satisfies $C_t H^1$. Thus Lemma 2.2 applies in either domain. Derivatives of (5) are finite sums of products of derivatives of U , proving the stated bounds and support. For mean zero use $(U \cdot \nabla)U = \text{div}(U \otimes U)$ and integrate. The effective cutoff formulas and differentiation give the force program. Periodic evaluation uses a finite superset of potentially active translates, including adjoining intervals at a seam. $\square$

Two refinements of the whole-space comparison will be used explicitly. Lemma 2.2(ii) permits pressure in H^1 modulo spatial constants at each time, without continuity into that norm, when the competing velocity retains $C_t H^2 \cap C_t^1 L^2$ and finite-interval bounds on velocity and gradient. Its case (i) instead permits omission of the competing-gradient bound when pressure belongs to $C_t H^1$. The compactly supported reference fields below satisfy the hypotheses of both statements. We retain each theorem's declared comparison class when applying the corresponding case.

For each construction below, the finite rule list gives all spatial branches before a trajectory is chosen. The choice of initial coordinates or a loading motion supplies the exact rational starting code; a loading rule inside the repeated program is included in the induction at integer times. We then check the entire path of every used nonhalting branch against its observer. This last check is essential: a safe pair of endpoints alone need not give a safe interpolation. Bounds on the finitely many pulse types give uniform kinetic energy on the torus, or under fixed compact support in Euclidean space. No simulation trace enters a force prescription.

4 The first initialized simulation

We now use the moving-sheet construction to prove the first result. Records interspersed among work symbols retain the information erased by a machine instruction. The processor below first supplies its exact initialized simulation and the stronger separation of its full rule cylinders. We then place those cylinders and check the fixed observer during loading and throughout every subsequent motion.

4.1 History interspersed among work symbols

Let Γ be the work alphabet, Q the states, and H the halt set. Adjoin an origin bit to each work symbol, preserving it under writes; initially only cell zero has bit one. Write $A = \Gamma \times \{0, 1\}$ and $b_0 = (\square, 0)$. For each nonhalt instruction $\hat{\delta}(q, a) = (q', b, d)$, $d \in \{0, +, -\}$, and incoming mode $h \in \{0, +, -\}$, introduce a distinct record $g_{q,h,a}$. The records form an alphabet G disjoint from A . Main controls are $M(q, h)$; $C_+(q)$, $C_-(q)$ finish a head movement.

Lemma 4.1 (Interspersed-history simulation). *Starting at $M(q_0, 0)$ with left stack b_0^∞ and right stack the origin-tagged input followed by blanks, this table simulates every ordinary step in finitely many positive rule applications. Erasing records at a main control gives the work tape, head, and state. The origin bit gives absolute tape positions. The full source cylinders are pairwise disjoint, as are the full image cylinders. The initialized run reaches a halting main control exactly when the machine halts.*

Proof. At a main control, erasing records makes R the head-and-right tape and L the reversed left tape. The first row writes b and saves the lost information in g . A stay is complete. For a right move, b has been put on L and C_+ transfers the intervening records until the next work symbol is exposed. For a left move, C_- transfers the intervening records from L and finally its first work symbol; that symbol becomes the new head immediately before b after erasure. At finite rule time there are finitely many records and infinitely many work cells in each direction, so every scan finishes. Writes preserve the origin bit.

The complete rule list is

$$\begin{aligned}
 (M(q, h); \epsilon, a) &\rightarrow \begin{cases} (M(q', 0); g, b), & d = 0, \\ (C_+(q'); bg, \epsilon), & d = +, \quad g = g_{q,h,a}, \\ (C_-(q'); \epsilon, bg), & d = -, \end{cases} \\
 (C_+(q); \epsilon, g) &\rightarrow (C_+(q); g, \epsilon), & g \in G, \\
 (C_+(q); \epsilon, c) &\rightarrow (M(q, +); \epsilon, c), & c \in A, \\
 (C_-(q); g, \epsilon) &\rightarrow (C_-(q); \epsilon, g), & g \in G, \\
 (C_-(q); c, \epsilon) &\rightarrow (M(q, -); \epsilon, c), & c \in A.
 \end{aligned} \tag{6}$$

No rule leaves a main control with $q \in H$.

Source separation uses the tested top symbol on R for main and C_+ controls, and on L for C_- . Into C_+ the left prefix starts with a work symbol in a main entry and a record in a loop; the record names separate the main entries. Into C_- the same distinction is on R . Into $M(q, 0)$ the first left record identifies the incoming instruction; into $M(q, +)$ or $M(q, -)$ the first right work symbol identifies the exit. Thus every image has a unique inverse on its full cylinder. The simulation invariant proves the halt equivalence, including an initial halt. $\square$

We now prove Theorem 1.1.

Proof. For $K = |A \sqcup G|$, use digits $1, 3, \dots, 2K - 1$ and base $B = 2K + 2$. Enumerate all m controls by $i = 1, \dots, m$, put $\ell = 1/\lceil 16(m + 1) \rceil$, and use the code

$$(\beta_i + \ell E(L), 1/4 + \ell E(R), 1/4), \quad \beta_i = 2i\ell + \begin{cases} 5/8, & \text{halting,} \\ 1/4, & \text{otherwise.} \end{cases}$$

Nonhalt intervals lie in $(1/4, 3/8)$ and halt intervals in $(5/8, 3/4)$. All prefixes in (6) have length at most two; Lemma 3.1 therefore gives source and target gaps at least ℓ/B^2 . Let N be the number of rules. Apply Lemma 3.3, with collar at most $\min(\ell/(4B^2), 1/\lceil 16(N + 1) \rceil, 1/16)$, to their full closed rectangles. Its normal scale compensates the unequal total prefix lengths. The initial code is rational. A localized curl translation of radius $1/32$ loads it along the segment from the fixed label during $[0, 1]$. Repeat the unit rule pulse thereafter. All pulses have zero collars inside the cube and are curls, giving a smooth mean-zero velocity U with $U(0) = 0$. Lemma 3.4 supplies the force.

At time $1 + k$ induction gives the code after k rules until the first halt. A halt is in the observed arc, including at $t = 1$ if initial. For a nonhalting run the loader stays below $1/2$, and every subsequent horizontal coordinate is constant during vertical motion and between its two nonhalt endpoints during the middle stage. Thus the observer is avoided at every time. At rule time k the raw stacks are blank beyond $\max(1, |w|) + 2k$ symbols, so this code retains the whole configuration, not merely the event bit. $\square$

5 Prescribing a positive thickness

We shall also need boxes of a prescribed positive thickness, for which an unbounded choice of layer heights is convenient.

Lemma 5.1 (Exponential scaling of full boxes). *Suppose $R_i^\pm \subset \mathbb{R}^2$, $1 \leq i \leq N$, $N \geq 1$, are separately separated nondegenerate rational closed rectangles. Let F_i be their positive diagonal affine maps, and explicitly require $F_i(R_i^-) = R_i^+$. Their factors $\lambda_{i1}, \lambda_{i2}$ are positive rational numbers, given by the corresponding rational side-length ratios. Put $\lambda_{i3} = (\lambda_{i1}\lambda_{i2})^{-1}$, and choose any positive rational source half-heights h_i^- . Define $h_i^+ = \lambda_{i3}h_i^-$. There is an effective compactly supported divergence-free unit-time pulse whose map on each full box $R_i^- \times [-h_i^-, h_i^-]$ is the specified horizontal map together with $z \mapsto \lambda_{i3}z$. It maps that box to $R_i^+ \times [-h_i^+, h_i^+]$, and the formula is exact on a positive three-dimensional neighborhood. Its centers interpolate linearly in the horizontal coordinates, while every half-side is at most the larger of its endpoint half-sides. The pulse vanishes near both time endpoints.*

Proof. Let $Z = \max_{i,\pm} h_i^\pm$. Choose a positive rational collar $0 < \eta \leq 1/4$ no larger than one quarter of each within-family horizontal coordinate-separating gap. If a family has only one member, there is no pairwise condition. Assign heights $z_i = 4(Z + 1)i$. More generally, the same proof permits any heights whose pairwise gaps exceed $2Z + 2\eta$. With successive flat switches s_1, s_2, s_3 , lift, deform, and lower using

$$C_i = ((1 - s_2)c_i^- + s_2c_i^+, z_i(s_1 - s_3)), \quad D_i = \text{diag}(\lambda_{i1}^{s_2}, \lambda_{i2}^{s_2}, \lambda_{i3}^{s_2}).$$

The middle-stage vertical half-size is at most Z , because every positive geometric interpolation stays between its two endpoint values. During ascent and descent, the fixed source and target projections, respectively, keep the η enlargements disjoint. During deformation the height condition separates those enlargements. Product cutoffs equal to one on the $\eta/2$ enlargement and supported in the η enlargement therefore meet Lemma 3.2. The determinant of D_i is one. That lemma gives the exact

affine path on every full box and a positive initial neighborhood. Only finitely many bounded paths occur, so the support lies in a fixed compact set. Logarithms and exponentials of positive rational factors are effective.

This proof permits different initial thicknesses because it uses only the finite maximum of all endpoint half-heights. In the special case $h_i^- = 1$ one has $Z = \max_i(1, \lambda_{i3})$ and the stated default heights are $4(Z + 1)i$. Unlike the linear-width sheet path, this interpolation does not make each individual horizontal trajectory a convex combination of its endpoints; the center and half-width bounds are the assertions used below. $\square$

5.1 Records attached to individual work cells

A related recorder keeps a word of history immediately after each work cell. This changes the actual rule cylinders: a stay writes its record on the right, and a right move transfers its newly written history in a separate scan. It is not a renaming of (6).

Use an origin-tagged work alphabet Σ , moves $D = \{0, +, -\}$, and an incoming tag $e \in D \sqcup \{*\}$. Extend halt states by idle instructions. The controls are $C(q, e), F(p), G(p)$; the distinct records are $\gamma(q, e, a)$, all outside Σ ; write $\mathcal{G}$ for their alphabet. Initialize at $C(q_0, *)$ with stacks $\square^\infty$ and $w\square^\infty$, where w includes the permanent origin tag. At a main control with head at j , its invariant is

$$L = \text{rev}(a_{j-1}W_{j-1}) \text{rev}(a_{j-2}W_{j-2}) \cdots, \quad R = a_jW_j a_{j+1}W_{j+1} \cdots,$$

where every W_i is a finite record word, initially empty. For an instruction writing b and retaining its record g , the main update changes a_jW_j to bgW_j . The F scan transfers the whole record word gW_j before exposing the next work cell; the G scan transfers the previous cell's reversed record and then its work symbol. Thus every instruction finishes in finitely many positive steps and gives the stated invariant. Erasing records and using the origin tag recovers the original tape.

If $\delta(q, a) = (p, b, d)$ and $g = \gamma(q, e, a)$, use

$$\begin{aligned} d = 0 : & \quad (C(q, e); \epsilon, a) \rightarrow (C(p, 0); \epsilon, bg), \\ d = + : & \quad (C(q, e); \epsilon, a) \rightarrow (F(p); b, g), \\ d = - : & \quad (C(q, e); \epsilon, a) \rightarrow (G(p); \epsilon, bg). \end{aligned} \tag{7}$$

For each record g' and work symbol a' , add

$$\begin{aligned} (F(p); \epsilon, g') & \rightarrow (F(p); g', \epsilon), & (F(p); \epsilon, a') & \rightarrow (C(p, +); \epsilon, a'), \\ (G(p); g', \epsilon) & \rightarrow (G(p); \epsilon, g'), & (G(p); a', \epsilon) & \rightarrow (C(p, -); \epsilon, a'). \end{aligned} \tag{8}$$

The sources are separated by their tested top symbols. Image cylinders into $C(p, 0)$ are separated by their right prefixes bg , and those into $C(p, +)$ or $C(p, -)$ by the exiting work symbol. No rule enters $C(p, *)$. Into $F(p)$ a main entry has work first on L and its identifying record first on R , whereas a loop has a record first on L . Into $G(p)$ a main entry has bg first on R , whereas a loop has a record first there. These distinctions recover the incoming rule for all tails.

Theorem 5.2 (Input shifts and periodic full-box motion). *This recorder gives an effective smooth general force on $\mathbb{R}^3$ with fixed compact spatial support, period one from $t = 0$, and all mixed derivatives bounded. Its zero-data solution is globally smooth with uniformly bounded energy. The particle starting at the origin has first coordinate greater than 2 at some time exactly when the machine halts. Uniqueness holds for smooth velocities in $C_t H^2 \cap C_t^1 L^2$ with velocity and gradient bounded on finite intervals, and pressure in H^1 modulo constants at each time. Pressure is determined modulo a function of time.*

Proof. Take odd digits and base $B = 2|\Sigma \sqcup \mathcal{G}| + 1$, where $\mathcal{G}$ is the record alphabet. Put $l_0 = E(\square^\infty)$ and $r_0 = E(w\square^\infty)$. Give $C(q_0, *)$ offset zero, other nontrigger controls distinct offsets $-4, -8, \dots$, and trigger controls $C(q, d)$ with q halting and $d \in D$ offsets $4, 8, \dots$. Encode by

$$(o_s - l_0 + E(L), -r_0 + E(R), 0). \quad (9)$$

The initial code is exactly zero. For every rule use its two prefix rectangles, with scales $\lambda_1 = B^{|v|-|v'|}$, $\lambda_2 = B^{|w|-|w'|}$ and $\lambda_3 = (\lambda_1\lambda_2)^{-1}$. Lemma 5.1 acts on the entire box $R^- \times [-1, 1]$, sending it to $R^+ \times [-\lambda_3, \lambda_3]$. The encoded central sheet returns to the same plane $z = 0$ on successive periods; the full box thickness generally changes. One may take margin $B^{-m_*}/10$, where m_* is the maximum prefix length, and heights $4(Z+1)i$ as in that lemma. Periodize the resulting pulse from time zero; the time collars give $U(0) = 0$. Lemma 3.4 and its pointwise H^1 pressure refinement give the equation and uniqueness.

The code induction now begins at $t = 0$. A reached halt has a trigger code with first coordinate at least $4 - l_0 > 2$. An initial halt reaches one after its first idle instruction, changing tag $*$ to 0. For a nonhalting run, every used source and target has first-coordinate center at most 1 and half-width at most $1/2$. The exponential box path has convexly interpolated centers and half-width at most the larger endpoint half-width; hence it stays below $3/2 < 2$ throughout. Input shifts affect the finite spatial prescription, and no loader is required. $\square$

6 Planar routing and invariant coding planes

The contraction, parking and expansion strategy has an area-preserving predecessor in [3, Proposition 5.1]. The proof below gives the needed planar motion explicitly; the three-dimensional incompressibility is supplied separately by the invariant-plane lift.

Lemma 6.1 (Planar routing through parking positions). *Let D be an open convex rational rectangle. Let B_i and C_i be two separately disjoint finite families of nondegenerate rational closed rectangles compactly contained in D . An effective smooth planar field, supported in $(0, 1) \times D$, realizes their matched positive diagonal affine maps on positive neighborhoods of all B_i . If $G \subset D$ is an open convex rational rectangle, every pair $B_i, C_i \in G$ can have its entire controlled motion confined to G .*

Proof. Take $G = D$ if no smaller safe rectangle is specified. For an empty family use zero. Otherwise choose distinct rational parking points in G avoiding all source and target centers. Move centers first from the sources to their parking points, one at a time, and then from parking to their targets, one at a time. Each destination is unoccupied. For a move $p \rightarrow q$, a two-segment path $p \rightarrow a \rightarrow q$ avoids the finite set of stationary centers if a avoids the lines joining an endpoint to one of those centers. Choose a rational in G when both endpoints are there, and in D otherwise. To make this choice effective, take a rational parabola segment in the chosen open rectangle: every forbidden line meets it at at most two points, so finitely many rational samples suffice. Convexity keeps the segments inside the chosen rectangle.

All segment-to-stationary-center squared distances and boundary clearances have positive rational lower bounds. Choose a rational $\epsilon > 0$ smaller than every endpoint half-side, with 10ϵ below every relevant distance and boundary clearance. Shrink each source rectangle to the square of half-side ϵ about its center, translate these squares along the chosen segments in order, and expand inside the target rectangles. During contraction the source cutoffs have pairwise disjoint supports; during expansion the target cutoffs do. The moving square's support of half-side 2ϵ misses each stationary square. During a shrink/expansion about c with positive half-widths $v_a(t)$, use a cutoff times

$$\left(\frac{\dot{v}_1}{v_1}(x_1 - c_1), \frac{\dot{v}_2}{v_2}(x_2 - c_2) \right);$$

during a translation use a moving plateau times $\dot{c}$. The exact path is $c(t) + \text{diag}(v_1(t), v_2(t))y$, $y \in [-1, 1]^2$. Its endpoint factors are the ratios of the target to source half-sides. Flat schedules on finitely many successive slots make a smooth pulse. Shrinking and expanding stay in the endpoint rectangles; the prescribed path clearances prove the extra confinement in G .

Widths may be interpolated exponentially or linearly. With padded plateaus, either choice gives an effective positive initial neighborhood by the finite products of affine norms. A cutoff that equals one only on the closed moving rectangle still proves the same formula there, including its boundary, but by itself asserts no exterior neighborhood. We shall state explicitly when such a closed-rectangle implementation is being retained. The empty-family case uses the zero field. $\square$

Lemma 6.2 (Invariant-plane lift). *Let $b(t, r)$ be an effective smooth planar field, periodic on $\mathbb{T}^2$ or compactly supported in $\mathbb{R}^2$, with bounded mixed derivatives. Let h be smooth effective with bounded derivatives, $h(z_0) = 0$ and $h'(z_0) = 1$. Then*

$$U(t, r, z) = (h'(z)b_1, h'(z)b_2, -h(z) \text{div}_r b) \quad (10)$$

is divergence free and carries exactly the planar flow on $z = z_0$. The choice $h(z) = \sin(2\pi z)/(2\pi)$ for $z_0 = 0$ gives a mean-zero unit-torus field. The choice $h = (z - z_0)\chi(z)$, with $\chi = 1$ near z_0 and compact support, gives compact support on $\mathbb{R}^3$ when b is compactly supported, or a mean-zero torus field when h is supported in one vertical chart. Time periods and mixed-derivative bounds are preserved.

Proof. The divergence is $h' \text{div} b - h' \text{div} b = 0$. On the indicated plane $U = (b, 0)$, so ODE uniqueness gives its invariance. For the sine choice both vertical factors integrate to zero. For compact h , $\int h' = 0$ and $\int \text{div} b = 0$ give the three means. All remaining assertions follow by differentiation of finite products. $\square$

7 Histories above and below separators

A separator lets one stack contain both work symbols and a finite history without mixing their roles. Storing history below the separator makes the scan length depend on the updated work word. Storing it above the separator makes the scan traverse the accumulated history instead. This choice changes the instruction count and the unused image cylinders available for initialization.

7.1 History below a delimiter

A delimiter lets a finite left work word sit above its history. A marker on the right remembers how far the subsequent return scan must go. Use work alphabet Γ and normalize to one halt state q_h . Introduce $\#$, $\star$ and records h_i . For a right or stay instruction $\delta(q, a) = (q', b, d)$ let $i = (q, a)$; for a left instruction let $i = (q, a, c)$ with $c \in \Gamma \sqcup \{\#\}$. Write $q^+(i) = q'$.

Lemma 7.1 (Delimiter-history simulation). *Starting at $(N_{q_0}, \# \square^\infty, w \square^\infty)$, normal configurations have form $(N_q, L \# H \square^\infty, R)$, where L, H are finite, R is eventually blank, and the represented left tape is $L \square^\infty$. One machine instruction takes exactly $2|L'| + 3$ rules, where L' is the updated finite left work word. Full source and image cylinders are separately disjoint, and the initialized run reaches N_{q_h} exactly when the machine halts.*

Proof. For $R = aR_*$ the first rule gives the updated pair

$$(L', R') = \begin{cases} (bL, R_*), & d = +, \\ (L, bR_*), & d = 0, \\ (L_*, cbR_*), & d = -, L = cL_*, \\ (\epsilon, \square bR_*), & d = -, L = \epsilon. \end{cases}$$

It also places $\star$ before R' . The D_i scan transfers $|L'|$ work symbols until $\#$ is exposed, inserts h_i below it, and enters $U_{q^+(i)}$. Exactly $|L'|$ reverse transfers restore L' , and the marker-removal rule finishes the step. This proves the count and tape invariant, including the new blank at the left boundary. Record displacements recover the absolute head position.

The complete rules are

$$\begin{aligned} (N_q; \epsilon, a) &\rightarrow (D_i; b, \star), & d = +, \\ (N_q; \epsilon, a) &\rightarrow (D_i; \epsilon, \star b), & d = 0, \\ (N_q; c, a) &\rightarrow (D_i; \epsilon, \star cb), & d = -, c \in \Gamma, \\ (N_q; \#, a) &\rightarrow (D_i; \#, \star \square b), & d = -, c = \#, \\ (D_i; c, \epsilon) &\rightarrow (D_i; \epsilon, c), & c \in \Gamma, \\ (D_i; \#, \epsilon) &\rightarrow (U_{q^+(i)}; \#h_i, \epsilon), \\ (U_q; \epsilon, c) &\rightarrow (U_q; c, \epsilon), & c \in \Gamma, \\ (U_q; \epsilon, \star) &\rightarrow (N_q; \epsilon, \epsilon). \end{aligned} \tag{11}$$

No normal rule leaves N_{q_h} .

The tested top symbols separate sources, with an extra left-top test for left moves. Into D_i a normal entry starts with $\star$ on the right and a loop with an ordinary letter. Into U_q a history entry starts with $\#h_i$ on the left and a loop with an ordinary letter. Distinct records identify distinct entries. Into N_q there is just its marker-removal rule. These are inverse checks for arbitrary tails, so induction gives all assertions including initial halting. $\square$

Three useful choices of observation follow from this same delimiter mechanism. One uses its table directly; a second changes its coordinates; a third separates the work update and marker push into two genuine rules. This last modification changes the sample times even though contracting those two rules recovers the original table.

Theorem 7.2 (Delimiter processors on invariant planes). *For every machine and finite input, each row below gives an effective smooth mean-zero general force on the unit torus, periodic of period one for $t \geq 1$, with all mixed derivatives bounded. Its zero-data solution is unique in the classical periodic class and has uniformly bounded energy. The stated fixed particle enters the stated fixed open set exactly when the machine halts.*

Processor	Initial label	Open set
Delimiter	$(1/2, 1/2, 0)$	$(2/3, 5/6)^2 \times \mathbb{T}$
Delimiter, origin coordinates	$(0, 0, 0)$	$\{x : 2/3 < x_1 < 11/12\}$
Separate marker push	$(1/4, 1/4, 0)$	$\{x : 3/4 < x_1 < 1\}$

The repeated pulse uses only the machine; a separate loader uses the input. All formulas allow arbitrary fixed positive real viscosity as a parameter; absolute computability uses computable viscosity.

Proof. The third processor. For a branch i of (11), replace its normal rule by a rule that performs the same tape update but enters a fresh state B_i without pushing $\star$, followed by $(B_i; \epsilon, \epsilon) \rightarrow (D_i; \epsilon, \star)$. Use exactly the other rows of that table. Every B_i has one incoming and one outgoing rule. Into D_i its new entry has right top $\star$, whereas loop entries have ordinary right top, so the previous full-cylinder inverse proof remains valid. On every full source cylinder, the composite of the two new rules equals the old normal rule. This proves the exact relationship, and the machine-step count is now $2|L'| + 4$.

Codes and placement. Use odd digits and base $B = 2m + 1$, where m is the relevant full alphabet size. Apply Lemma 3.1 in the following state rectangles. For the first row take the halt square $[7/10, 4/5]^2$. For the M other controls take centers $(1/8 + (2j + 1)/(8M), 1/4)$, $0 \leq j < M$, and half-side $1/(16M)$. For the second row take the halt rectangle $[3/4, 7/8] \times [1/4, 1/2]$ and other rectangles

$$[1/8 + j/(4M), 1/8 + j/(4M) + 1/(8M)] \times [1/4, 1/2].$$

For the third row give the halt control the square centered at $(7/8, 1/2)$ of side $1/16$, and the other M controls squares centered at $(i/[2(M + 1)], 1/2)$, $1 \leq i \leq M$, of side $1/[8(M + 1)]$. The first two nonhalt families lie in $(0, 1/2)^2$, respectively $(0, 1/2) \times (0, 1)$; the third lies in the latter open rectangle. In each case the full source and target rectangle families are separately separated by the proved cylinder checks and digit gaps.

Routing and loading. Apply Lemma 6.1 in $(0, 1)^2$, using the corresponding nonhalt open rectangle as the safe region. This proves exact maps on full rectangles and all-time confinement whenever both endpoints are nonhalting. Both its exponential-width and linear-width implementations are available. The second row also permits the following explicit linear-width choice: park strictly to the left of all endpoint centers and choose every bend to the left of both endpoints, off the finite forbidden lines. Nonhalt centers then remain below $3/8$; small squares of half-side below $1/8$ stay below $1/2$. This gives the same endpoint maps with a directly checked closed-rectangle plateau. For the third row one may choose rational bends on $(\lambda s, s^2)$, $\lambda = 1/2$ or 1 . After clearing denominators, forbidden lines give nonzero integer polynomials of degree at most two. A rational $s = 1/D$ with D larger than all absolute leading coefficients cannot be a root by the rational-root theorem. Avoid the finite endpoint denominators as well and take D large enough for the prescribed open rectangle. This is an effective form of the parking choice, with no generic-position assumption.

All initial codes are rational. In the first row a planar compact translation, with plateau half-side $1/100$ and support half-side $2/100$, loads from $(1/2, 1/2)$. In the second row the spatially constant planar velocity $\dot{\theta}(t)r_{\text{in}}$ loads from the origin. In the third row a small compact translation tube loads from $(1/4, 1/4)$. The loader paths avoid the respective observers when the initial state is nonhalting. Follow the loader by the repeated unit pulse and apply Lemma 6.2 with $h = \sin(2\pi z)/(2\pi)$. The three velocities are mean zero, smooth at all joins, and zero initially; Lemma 3.4 supplies their forces.

At time $1 + k$ the particle is the code after k primitive rules until halting. A halt code lies strictly in the corresponding observed set, including an initial halt at time one. For nonhalting runs, the loader and every subsequent controlled path avoid that set by the established safe-region bounds. This proves the continuous-time equivalences and all three force contracts. $\square$

7.2 An input instruction inside the repeating program

A fixed initial code can instead load the input through an ordinary prefix rule. This makes the whole force periodic from zero, at the cost of placing the input in its repeating finite table. Use a

new initial state S , and the delimiter alphabet renamed $\sharp, \diamond, h_j$. The input rule is

$$(S; \epsilon, \epsilon) \rightarrow (F_{j*}; \epsilon, \diamond w), \quad q(j_*) = q_0.$$

For each nonhalt instruction $\delta(q, a) = (q', b, d)$ use primary rules

$$\begin{aligned} d = + : & \quad (q; \epsilon, a) \rightarrow (F_j; b, \diamond), \\ d = 0 : & \quad (q; \epsilon, a) \rightarrow (F_j; \epsilon, \diamond b), \\ d = - : & \quad (q; c, a) \rightarrow (F_j; \epsilon, \diamond cb), \quad c \in \Gamma, \\ & \quad (q; \sharp, a) \rightarrow (F_j; \sharp, \diamond \square b). \end{aligned}$$

Every alternative, including initialization, has its own j and history letter h_j , and $q(j) = q'$. Complete the table by

$$\begin{aligned} (F_j; c, \epsilon) &\rightarrow (F_j; \epsilon, c), & c \in \Gamma, \\ (F_j; \sharp, \epsilon) &\rightarrow (G_{q(j)}; \sharp h_j, \epsilon), \\ (G_q; \epsilon, c) &\rightarrow (G_q; c, \epsilon), & c \in \Gamma, \\ (G_q; \epsilon, \diamond) &\rightarrow (q; \epsilon, \epsilon). \end{aligned} \tag{12}$$

There are no rules from halt controls or into S .

Lemma 7.3. *These source and image cylinders are separately disjoint. From $(S, \sharp \square^\infty, \square^\infty)$ the first procedure inserts w , and thereafter one ordinary step takes $2|L'| + 3$ rules and reaches the correct main configuration. The run reaches a halt main control exactly when the machine halts.*

Proof. Every F_j has a unique primary entry with right top $\diamond$; its loops have ordinary right tops. Entries into G_q have distinct left prefixes $\sharp h_j$, or ordinary left tops for loops. Each main state has its unique exit from G_q . Source separation is by the same tested top symbols as in Lemma 7.1, with the single additional source state S . These checks prove the full-domain claim. At main states the stacks are $(L \sharp H \square^\infty, R)$. The primary update gives $(L' \sharp H \square^\infty, \diamond R')$; the two scans transfer and restore L' , insert h_j , and delete the marker. Their count and interpretation are those proved for the delimiter table. Initialization is its case $L' = \epsilon$, $R' = w \square^\infty$ and takes three rules. The history records include the zero-displacement initialization and recover the absolute head. $\square$

Theorem 7.4 (Input insertion in a periodic compact force). *This processor gives an effective smooth general force on $\mathbb{R}^3$ with period one from zero, a fixed compact spatial support, and all mixed derivatives bounded. Its zero-data solution is globally smooth, has uniformly bounded energy, and is unique among smooth classical solutions with $u \in C_t H^2 \cap C_t^1 L^2$, $p \in C_t H^1$, and bounded velocity and gradient on each finite time interval. The particle from the origin has first coordinate greater than 2 at some time exactly when the machine halts. The formula is effective relative to any fixed real $\nu > 0$.*

Proof. For m letters use digits $2, 4, \dots, 2m$ and base $K = 2m + 2$. Set $a_S = -E(\sharp \square^\infty)$, give halt main states distinct offsets $4, 8, \dots$, and other states offsets $-4, -8, \dots$. With $b_0 = -E(\square^\infty)$, encode by $(a_s + E(L), b_0 + E(R), 0)$, so the fixed initial code is zero. Lemma 3.1 gives separated source and target rectangles. Thicken to $R_i^- \times [-1, 1]$ and apply Lemma 5.1 with scales $K^{|v|-|v'|}, K^{|w|-|w'|}$ and their reciprocal product. For example take the three switches $\sigma(8t - (2j - 1))$, $j = 1, 2, 3$, and collar one quarter of the minimum of 1 and the rectangle gaps. The field is zero for $t \leq 1/8$ and $t \geq 6/8$ within a period. Its periodization is smooth from zero, so Lemma 3.4 applies.

Induction at integer times follows the initialized rule sequence. A halt code has first coordinate at least 4. Every nonhalting used branch has endpoint centers at most 1 and widths at most 1; the exponential-width bound gives first coordinate at most $3/2$ throughout. The force therefore has the required all-time event. The complete input-insertion branch is included in every period, and its actual one-time use follows from the symbolic dynamics. $\square$

7.3 A finite work window on each side

The preceding delimiter processor leaves the right tape infinite. If both work words are finite, crossing either end must instead insert a new blank. This changes the guards of the right-move rules and introduces a distinct marker-push state. Here the work alphabet is Γ and the sole halt state is denoted by h .

At a ready control A_q use stacks $L\#H\Box^\infty$ and $R\#\Box^\infty$, where L, H are finite and R is a nonempty work word. Initially $L = H = \epsilon$ and $R = w$, except that empty input is replaced by the one-letter word $\Box$. Denote this initial right work word by R_{in} . For $\delta(q, a) = (q', b, d)$ the work prefix replacements are

d	(v, w)	(v', w')	condition
0	(ϵ, a)	(ϵ, b)	$c \in \Gamma$
+	(ϵ, ac)	(b, c)	
+	$(\epsilon, a\#)$	$(b, \Box\#)$	
−	(c, a)	(ϵ, cb)	$c \in \Gamma$
−	$(\#, a)$	$(\#, \Box b)$	

Give every alternative its own index r , control P_r , and record ρ_r . Its state change is $A_q \rightarrow P_r$ and $q'(r) = q'$. Add a marker J and controls S_r, T_q , where q runs through every machine state, including h . The rules are

$$\begin{aligned}
(P_r; \epsilon, \epsilon) &\rightarrow (S_r; \epsilon, J), \\
(S_r; c, \epsilon) &\rightarrow (S_r; \epsilon, c), & c \in \Gamma, \\
(S_r; \#, \epsilon) &\rightarrow (T_{q'(r)}; \#\rho_r, \epsilon), \\
(T_q; \epsilon, c) &\rightarrow (T_q; c, \epsilon), & c \in \Gamma, \\
(T_q; \epsilon, J) &\rightarrow (A_q; \epsilon, \epsilon).
\end{aligned}$$

The full alphabet is $\Gamma_{\text{full}} = \Gamma \sqcup \{\#, J\} \sqcup \{\rho_r\}_r$, with all added symbols distinct. There is no work rule from A_h . The work rule updates the finite tape interval, adjoining a blank exactly when a move crosses one of its ends. After pushing J , the S_r scan transfers the updated left word, inserts ρ_r below its delimiter, and the T_q scan restores it and removes J . Hence one step takes $4 + 2|L'|$ rules and leaves R' nonempty. All finite scans terminate. Sources at A_q are separated by the scanned symbol and the displayed additional boundary test. Each P_r has one incoming rule. Images into S_r have right top J or an ordinary letter; images into T_q have left prefix $\#\rho_r$ or an ordinary letter. Each A_q has one marker-removal entry. These observations prove source and image separation for arbitrary tails, as well as the initialized halt equivalence.

Theorem 7.5 (Finite windows and a fixed slab). *This processor gives an effective smooth mean-zero force on the unit torus, periodic of period one for $t \geq 1$, with every mixed derivative bounded. Its zero-data solution is globally smooth, unique in the classical periodic class, and has uniformly bounded energy. The particle from $(1/4, 1/4, 1/4)$ enters $\{x : 1/2 < x_1 < 3/4\}$ exactly when the machine halts. A general residual force with pressure zero or a solenoidal projected force with its accompanying pressure may be chosen; both give the same velocity. The repeated pulse depends only on the table and the loader on the input.*

Proof. Let K be the number of controls, $\eta = 1/(16K)$, and use odd digits with base $B = 2|\Gamma_{\text{full}}| + 1$. Give nonhalt controls offsets $a_{s_j} = 1/4 + 2j\eta$ and set $a_{A_h} = 5/8$. The code is

$$(a_s + \eta E(L), 1/4 + \eta E(R), 1/4).$$

All nonhalt state intervals are in $[1/4, 3/8)$, while the halt interval lies in $(1/2, 3/4)$. Lemma 3.1 gives separately separated rectangles $E_i^0, E_i^1 \subset [1/4, 3/4]^2$ and scale factors $\alpha_i = B^{|v_i^0| - |v_i^1|}$, $\beta_i = B^{|w_i^0| - |w_i^1|}$.

Write F_i for the positive diagonal affine map $E_i^0 \rightarrow E_i^1$. Here is an explicit thin-box instance of normal compensation. If there are m rules, put

$$Z_i = \frac{1}{2} + \frac{i}{4(m+1)}, \quad M = \max_i \max(1, 1/\alpha_i) \max(1, 1/\beta_i), \quad \delta_z = \frac{1}{64(m+1)M},$$

$$\mu = \min\{d_*/4, 1/[64(m+1)]\},$$

where d_* is the minimum of 1 and the within-family planar sup-norm distances. Lift the boxes $E_i^0 \times [1/4 - \delta_z, 1/4 + \delta_z]$ to Z_i , use the horizontal scales $1 - s + s\alpha_i$, $1 - s + s\beta_i$ and reciprocal vertical scale, and lower. Each vertical half-size is at most $\delta_z M$. During ascent/descent use the endpoint rectangle gaps; in the middle stage use the height gaps $1/[4(m+1)]$. Enlarging by μ preserves separation and stays in the cube. Thus Lemma 3.2 gives exact affine motion on the full boxes and positive collars. On their central sheet the first coordinate is $(1-s)X_1 + sF_i(X)_1$.

The input point is the code of $(A_{q_0}, \# \square^\infty, R_{\text{in}} \# \square^\infty)$. A curl translation with plateau half-side $1/32$ and margin $1/32$ loads it from the fixed label during $[0, 1]$. All paths stay inside the cube. Repetition gives a mean-zero velocity and Lemma 3.4 gives its residual force r and pressure zero. For the second choice, Proposition 2.4 solves $\Delta\psi = \text{div } r$, $\int \psi = 0$, and gives

$$f_{\text{sol}} = r - \nabla\psi, \quad p = -\psi.$$

It preserves all stated bounds and eventual periodicity, with precisely this change in pressure.

Induction at times $1+k$ gives the prefix run until halt. A halt is in the slab, including an initial halt. For a nonhalting run the loader stays below $1/2$ and each subsequent central-sheet coordinate remains between two nonhalt endpoints. No intermediate time enters the slab. The history recovers head displacement; the finite words above the two delimiters recover the full represented tape. $\square$

7.4 History above a separator and a repeated loader

If history is above the left tape separator, every instruction scans its entire current length. The number of primitive steps is then explicit. This also provides a free target cylinder in which a loading branch can be included in every period.

Use work alphabet Γ and normalize to one halt state q_H . Write the ordinary instructions as $j : (r_j, b_j^{\text{in}}) \mapsto (r_j^+, b_j^{\text{out}}, d_j)$. Adjoin a history alphabet $G = \{g_j\}_j$ and markers M_L, M_R , all disjoint from Γ . Let $I_j = \{\perp\}$ for $d_j \neq -1$, and $I_j = \Gamma$ for $d_j = -1$. Besides the main controls use $D_j, E_{j,c}$. Here $S_{j,c}$ is $\epsilon, b_j^{\text{out}}$, or cb_j^{out} for moves $1, 0, -1$, respectively.

At a main visit the stacks are $(HM_L L, R)$: L is the infinite reversed left tape, R starts at the head, and H is the finite history, newest first. Entry removes the scanned symbol and inserts M_R on the right. The D_j scan transfers H , then changes the left tape by pushing b_j^{out} , leaving it alone, or popping c , according to the move. The $E_{j,c}$ scan restores H and its exit gives $(g_j HM_L L', S_{j,c} R_*)$. This is exactly the next tape configuration and costs $2|H| + 3$ rules. Initially $H = \epsilon$, $L = \square^\infty$, $R = w\square^\infty$.

The complete rules are

$$\begin{aligned} (r_j; \epsilon, b_j^{\text{in}}) &\rightarrow (D_j; \epsilon, M_R), \\ (D_j; g, \epsilon) &\rightarrow (D_j; \epsilon, g), & g \in G, \\ (D_j; M_L, \epsilon) &\rightarrow (E_{j,\perp}; M_L b_j^{\text{out}}, \epsilon), & d_j = 1, \\ (D_j; M_L, \epsilon) &\rightarrow (E_{j,\perp}; M_L, \epsilon), & d_j = 0, \\ (D_j; M_L c, \epsilon) &\rightarrow (E_{j,c}; M_L, \epsilon), & d_j = -1, \quad c \in \Gamma, \\ (E_{j,c}; \epsilon, g) &\rightarrow (E_{j,c}; g, \epsilon), & g \in G, \\ (E_{j,c}; \epsilon, M_R) &\rightarrow (r_j^+; g_j, S_{j,c}), & c \in I_j, \end{aligned}$$

No rule leaves q_H .

Sources are separated by the ordinary read symbol, history tops, markers, and the extra left-neighbor test. For images into D_j , the right top is M_R after entry and g after a loop. Into $E_{j,c}$ the left top is M_L after modification and g after restoration. Into a main control the new left record g_j identifies the instruction; for a left-moving instruction the first right symbol further identifies c . These checks prove full source and image separation. They also give the initialized tape invariant and its exact halt equivalence.

Theorem 7.6 (A periodic processor with quadratic sample times). *Every machine and finite input give an effective smooth mean-zero general force on the unit torus, periodic of period one from time zero. Force and velocity have all mixed derivatives bounded and vanish near integer times. Their spatial supports lie in one compact subset of the cube chart. The unique classical periodic zero-data solution is globally smooth with uniformly bounded energy. From the fixed label $(1/2, 1/2, 1/4)$, the event $3/4 < X_1(t) < 7/8$ occurs exactly when the machine halts. Whenever reached, the configuration after k ordinary steps is encoded at time $(k + 1)^2$.*

Proof. For the full alphabet of size m use odd digits and base $B = 2m + 1$. If there are m_s controls, put $s_0 = 1/[16(m_s + 1)]$. The halt square has corner $(13/16, 1/4)$ and side s_0 ; all other squares have corners $(1/8 + 2js_0, 1/4)$ in their enumeration. Denote each assigned corner by p_s . The nonhalting corners lie below first coordinate $1/4$. Encode at height $z_0 = 1/4$ by

$$(p_s + s_0(E(A), E(B)), z_0).$$

Lemma 3.1 supplies all rule rectangles and their exact positive diagonal maps.

Add a pair of equal-size loader squares centered at $(1/2, 1/2)$ and

$$y_{\text{in}} = p_{q_0} + s_0(E(M_L \square^\infty), E(w \square^\infty)).$$

Its common half-side is half the minimum of $1/16$ and the distances from y_{in} to the sides of $R_{\text{sep}} = p_{q_0} + s_0(I(M_L) \times [0, 1])$. These distances are positive, since infinite odd-digit codes lie strictly inside their prefix intervals. The loader source misses every state square. Its target lies in R_{sep} , whereas all ordinary targets in q_0 have a history letter first on the left. Digit gaps therefore keep this new target separate. This verifies injectivity of the enlarged finite geometric family; it is the reason the loader may repeat.

Apply Lemma 3.3 to this family, including the loader. More explicitly, if there are N pairs, choose heights $h_i = 1/2 + i/[4(N + 1)]$. Write F_i for the affine map of pair i . For its horizontal scale factors λ_{ia} let

$$\Lambda = \max(\{1\} \cup \{\lambda_{ia} : 1 \leq i \leq N, a = 1, 2\}), \quad \Lambda_z = \max_i [\min(1, \lambda_{i1}) \min(1, \lambda_{i2})]^{-1}.$$

If g is the minimum of 1, horizontal boundary clearances, and the positive within-family coordinate-separating gaps, take horizontal source padding $g/(10\Lambda)$ and vertical half-thickness $1/[40(N + 1)\Lambda_z]$. The linear horizontal scale $1 - s + s\lambda_{ia}$ and reciprocal normal scale keep endpoint padding at most $g/10$ and middle-stage half-thickness at most one tenth of the height gap. The moving supports are consequently separate inside the cube. Use ascent, transformation, descent switches on $[1/8, 2/8]$, $[3/8, 5/8]$, $[6/8, 7/8]$. The curl formula gives a mean-zero periodic velocity with zero collars at integer times. Its central-sheet horizontal path is

$$(1 - s)y + sF_i(y). \tag{13}$$

Lemma 3.4 supplies the force, with pressure zero.

The first period loads the input. Thereafter each period performs the next primitive rule. After r ordinary steps $|H| = r$, so the sample time after k steps is $1 + \sum_{r=0}^{k-1} (2r + 3) = (k + 1)^2$. A halt code lies in the observed arc, also when $k = 0$. If the machine does not halt, all endpoints after loading have first coordinate below $1/4$, while the initial one is $1/2$. Equation (13) excludes the observed arc at every intermediate time. History displacements recover the absolute head index, and removing H, M_L recovers the tape. All branch and loader data were prepared from finite input, with no run computed. $\square$

8 History tracks and return scans

A separate tape track can store history while the work symbols remain in their indexed cells. Both processors in this section mark the work head, travel to a rightward history frontier, and return before completing the original head move. One advances the frontier through a separate control; the other edits its neighboring cell directly, which requires a different inverse guard on the return scan.

8.1 A marked head and a separate log track

The first compiler uses a separate instruction to advance the history frontier before returning to the marked work cell.

Let A be the work alphabet, Q the states, H the halt set, and $D = \{-1, 0, 1\}$. For $e = (q, s, a) \in (Q \setminus H) \times D \times A$ write $(r_e, b_e, d_e) = \delta(q, a)$. The full tape alphabet is $\Sigma = A \times (\{\ell, \#\} \sqcup \{[e]\}) \times \{0, 1\}$, where the last component marks the return location. Use controls $S_{q,s}, G_e, T_{r,d}, L_{r,d}$.

Initialize at $S_{q_0,0}$, head zero, with the prescribed work tape, all marks zero, and log blank except for $\#$ at cell one. At the k th main visit, records occupy $1, \dots, k$, the frontier is $j = k + 1$, and the ordinary head satisfies $|i| \leq k$. Thus $i < j$. The first rule writes the work symbol and marks i , G_e reaches the frontier, T advances it by one, and L returns to the sole mark, clears it, and performs the original move d_e . Every scanned symbol satisfies its displayed guard. These finite scans establish the next main configuration and the invariant by induction. Halting is exactly arrival at an $S_{q,s}$ with $q \in H$; this includes an initial halt.

The following instructions have output order (control, written symbol, move):

$$\begin{aligned} (S_{q,s}, (a, h, 0)) &\mapsto (G_e, (b_e, h, 1), 1) & h \neq \#, \\ (G_e, (c, h, 0)) &\mapsto (G_e, (c, h, 0), 1) & h \neq \#, \\ (G_e, (c, \#, 0)) &\mapsto (T_{r,d_e}, (c, [e], 0), 1), \\ (T_{r,d}, (c, \ell, 0)) &\mapsto (L_{r,d}, (c, \#, 0), -1), \\ (L_{r,d}, (c, h, 0)) &\mapsto (L_{r,d}, (c, h, 0), -1) & h \neq \#, \\ (L_{r,d}, (c, h, 1)) &\mapsto (S_{r,d}, (c, h, 0), d) & h \neq \#. \end{aligned}$$

The first row is restricted to nonhalt q , and no other rules exist.

All incoming moves to $G_e, T_{r,d}$ are rightward, those to $L_{r,d}$ leftward, and those to $S_{r,d}$ equal d . Moreover target control and written whole symbol recover the source rule. At G_e , the entering rule writes mark one while its loops write zero, and e recovers the old work symbol and control. At T the record recovers e . At L an entry writes $\#$ and a loop writes a different history symbol. At S the sole type of entry restores a mark from one to zero. This proves injectivity on the full partial domain, not just on the initialized tape.

Theorem 8.1 (A delayed head move on an invariant plane). *The marked-head compiler gives an effective smooth mean-zero general force on the unit torus, with all mixed derivatives bounded and period one for $t \geq 1$. Its unique classical periodic zero-data solution is globally smooth with bounded*

energy. The particle from $(1/4, 1/4, 1/2)$ enters $(5/8, 7/8)^2 \times \mathbb{T}$ exactly when the original machine halts. Only its loader depends on the input.

Proof. Put $N = (1/8, 3/8)^2$, $J = (5/8, 7/8)^2$. Place disjoint rational state squares in N for nonhalt controls and in J for halt controls. For k controls assigned to $(a, a+g)^2$, centers $(a+ig/(k+1), a+g/2)$ and half-side $g/[4(k+1)]$ suffice. Let P_s map $[0, 1]^2$ to the state square. With base $B = 2|\Sigma| + 1$, odd digits and $h_\sigma(x) = (d(\sigma) + x)/B$, an instruction $(s, \sigma) \mapsto (s', \tau, d)$ has normalized maps

$$\begin{aligned} d = 1 : & \quad (x, h_\sigma(y)) \mapsto (h_\tau(x), y), \\ d = 0 : & \quad (x, h_\sigma(y)) \mapsto (x, h_\tau(y)), \\ d = -1 : & \quad (h_\lambda(x), h_\sigma(y)) \mapsto (x, h_\lambda(h_\tau(y))), \quad \lambda \in \Sigma. \end{aligned}$$

All free coordinates range over $[0, 1]$. Determinism and read symbols separate sources. At a common target control the incoming move is fixed; distinct written symbols separate targets in the left first digit for $d = 1$, the right first digit for $d = 0$, and the right pair (λ, τ) for $d = -1$. Hence the full target rectangles are separated as well. Composing with $P_s, P_{s'}$ preserves positivity of the diagonal maps even when state squares have different sizes.

Apply Lemma 6.1 in $(0, 1)^2$ with safe region N . To retain the linear-width alternative explicitly, shrink to a common small rational half-side r , use sequential parking routes, and linearly interpolate centers and widths with flat switches. If $0 < \beta < 1$ makes the rectangles enlarged by $1 + \beta$ pairwise disjoint within each endpoint family, the cutoff

$$\prod_{a=1}^2 \left[1 - \sigma \left(\frac{((x_a - c_a)/v_a)^2 - 1}{(1 + \beta)^2 - 1} \right) \right]$$

multiplied by $\dot{c} + \text{diag}(\dot{v}_a/v_a)(x - c)$ gives the exact closed rectangle path $c(t) + \text{diag}(v_a(t)/v_a(0))(x_0 - c(0))$. Taking r below one sixteenth of every relevant rational squared clearance and endpoint half-side makes the small moving supports miss the stationary ones. Nonhalt paths remain in N throughout. This formula proves exactness on closed rectangles; the padded version of Lemma 6.1 separately gives exterior neighborhoods.

Load the rational initial code from $(1/4, 1/4)$ by the same planar routing of a small square, confined to N when the initial state is nonhalting. Follow by the unit step pulse. Lift with $h(z) = (z - 1/2)\chi(z)$, where $\chi = 1$ for $|z - 1/2| \leq 1/8$ and is supported in $|z - 1/2| < 1/4$. Lemmas 6.2 and 3.4 give the mean-zero velocity and force with pressure zero. At unit samples after loading the particle is the successive code. Halt codes lie in J ; a nonhalting run remains in N throughout loading and all later motions. This proves the event. The log's leftmost nonblank cell remains absolute cell one, so the encoded tape also determines absolute head location. $\square$

8.2 A pointer that edits the neighboring cell

The previous compiler advances the frontier through a separate control. The next one advances it by editing the next cell directly. This saves a state, but its returning scan needs a different guard. We state that guard on the entire partial domain before considering initialized tapes.

Let A be the work alphabet and $D = \{-1, 0, 1\}$. Put

$$\mathcal{R} = (Q \setminus H) \times D \times A, \quad K = \{\perp, \text{top}\} \sqcup \{[r] : r \in \mathcal{R}\}, \quad \Gamma = A \times K \times \{0, 1\}.$$

Let $\mathcal{S}$ consist of the controls $C_{q,e}, I_{q,e}, O_r$, with halting controls $\mathcal{H} = \{C_{q,e} : q \in H\}$. Index a configuration's tape relative to its head by $w_i = (a_i, k_i, m_i)$. A move ℓ after edits means $w'_i = \tilde{w}_{i+\ell}$. Unmentioned components remain unchanged. There are exactly five kinds of rules:

1. At $C_{q,e}$, $q \notin H$, $m_0 = 0$, let $r = (q, e, a_0)$ and $\delta(q, a_0) = (q', b, d)$. Write $a_0 = b, m_0 = 1$, enter O_r , move 1.
2. At O_r , $m_0 = 0$, $k_0 \neq \text{top}$, make no edits and move 1, retaining the control.
3. At O_r , $m_0 = 0$, $k_0 = \text{top}$, $k_1 = \perp$, write $k_0 = [r], k_1 = \text{top}$, enter $I_{q',d}$, and move -1 . Here q', d are determined by r .
4. At $I_{q',d}$, $m_0 = 0$, $k_1 \neq \text{top}$, make no edits and move -1 , retaining the control.
5. At $I_{q',d}$, $m_0 = 1$, clear that mark, enter $C_{q',d}$, and move d .

Lemma 8.2 (Neighboring-cell recorder). *This partial map G is injective. Initialize it at $C_{q_0,0}$ with head zero, finitely specified work tape, marks zero, and blank history except for its pointer at absolute cell $a \in \{1, 2\}$. At its n th compute-state visit the work configuration is the machine's n th configuration, marks are zero, records occupy $a, \dots, a + n - 1$, and the pointer is at $T_n = a + n$. If h_n is the head location, the next machine step takes exactly $2(T_n - h_n) + 1$ applications of G . In particular, its initialized run reaches $\mathcal{H}$ exactly when the machine halts.*

Proof. An output in O_r came by a right move. Its mark at relative position -1 distinguishes the compute entry (one) from a loop (zero). In the first case r restores the old state, incoming tag, and work symbol; in the second there were no edits. An output in $I_{q',d}$ came by a left move. Its history symbol at relative position 2 is a pointer precisely for the entry from O_r ; the return-loop guard excludes that case. For this entry the record at relative position 1 identifies r and restores the old pair $(\text{top}, \perp)$. An output in $C_{q',d}$ has a unique predecessor type: undo move d and restore mark one. This proves injectivity on every actual rule output; it does not assert surjectivity or impose initialized-tape restrictions.

Since $|h_n| \leq n$, $T_n > h_n$. The compute rule marks h_n ; the outgoing scan reaches T_n , records the old data, advances the pointer, and returns from $T_n - 1$. Every return-scan cell is unmarked until the marked one, and its right neighbor is never the new pointer. Thus each guard holds. Clearing the mark and moving by d gives the next machine configuration, with $h_n + d \leq T_n < T_n + 1$. The count is $1 + (T_n - h_n - 1) + 1 + (T_n - h_n - 1) + 1$. It is at most $4n + 5$ for $a = 2$ and at most $4n + 3$ for $a = 1$. This proves the induction and includes initial halting. $\square$

The two initial pointer positions remain distinct: the $a = 2$ initialization uses two more local steps per ordinary instruction than $a = 1$, by the exact count in Lemma 8.2.

To encode G , enumerate every applicable control and full triple (v_{-1}, v_0, v_1) . These triples include every tested and edited cell and partition the domain into finitely many cylinders. The left and right source prefixes have lengths 1 and 2. If the move is ℓ and the edited triple is $\tilde{v}$, the target prefixes are

$$L' = (\tilde{v}_{\ell-j})_{1 \leq j \leq \ell+1}, \quad R' = (\tilde{v}_{\ell+j-1})_{1 \leq j \leq 2-\ell}. \quad (14)$$

The old free tails at indices $-2, -3, \dots$ and $2, 3, \dots$ remain unchanged and ordered. Thus each image is a full cylinder; distinct images are disjoint by injectivity. Lemma 3.1 consequently gives full separated source and target rectangles, with linear parts $\text{diag}(B^{-\ell}, B^{\ell})$ when state-square scales agree.

Theorem 8.3 (Two invariant-plane realizations). *The neighboring-cell recorder has the following effective zero-data forced Navier–Stokes realizations at fixed positive computable viscosity. Both have all mixed derivatives bounded, period one for $t \geq 1$, and uniformly bounded kinetic energy.*

1. On the unit torus, any tape differing from blank on a specified finite set of cells is allowed. The force is mean zero and general, and the solution is unique in the classical periodic class. The particle from $(1/16, 1/16, 0)$ enters $\{x : 1/2 < x_1 < 7/8\}$ exactly when the machine halts.
2. On $\mathbb{R}^3$, for a finite word starting at cell zero, the force is general with one compact spatial support uniform in time. The particle from the origin has negative first coordinate at some time exactly when the machine halts. Uniqueness holds for smooth solutions with bounded velocity on finite intervals and $u \in C_t H^2 \cap C_t^1 L^2$, $p \in C_t H^1$; no bound on the competing velocity gradient is required.

The repeated field depends only on the table; only the loader uses the input. Neither force is required to decay or be solenoidal. The same formulas are effective relative to arbitrary fixed real positive viscosity.

Proof. For the torus use $a = 2$, odd digits, base $B = 2|\Gamma| + 1$, and $\lambda = 1/(16|\mathcal{S}|)$. Enumerating each type from zero, assign corners

$$b_s = (1/8 + 2j\lambda, 1/4) \quad (s \notin \mathcal{H}), \quad b_s = (5/8 + 2j\lambda, 1/4) \quad (s \in \mathcal{H}).$$

Use the code $b_s + \lambda(E(L), E(R))$. The nonhalt squares lie in $0 < x < 1/4$ and the halt squares in $1/2 < x < 7/8$. For Euclidean space use $a = 1$, base $B = 2|\Gamma|$ and digits $0, 2, \dots, B - 2$. Give halt controls offsets $-2, -4, \dots$ and all others offsets $2, 4, \dots$, writing $H(s)$ for the offset. Use $(H(s) + E(L), E(R))$. Zero digits can put codes on rectangle boundaries; our maps are exact on the full closed rectangles. Nonhalt rectangles lie in $x \geq 2$, and halt codes have $x < 0$. In each construction the prefixes in (14) give the separated rectangles required for planar routing.

Here are explicit choices retaining the two all-time avoidance bounds. Let N be the number of rectangle pairs. In the torus chart park at

$$p_i = (x_*, i/(N + 1)), \quad x_* = \frac{1}{2} \min(\{1/4\} \cup \{c_{i1}, d_{i1}\}),$$

where c_i, d_i are endpoint centers. A bend with $0 < x < 1/4$ can be chosen off the forbidden lines as follows: choose its abscissa below all vertical forbidden abscissae and $1/4$, then its positive ordinate below 1 and below all positive values of the other lines there. All segments stay in the chart; when both endpoints are nonhalt their centers stay below $1/4$. Let ρ be the minimum of 1, endpoint half-sides, boundary clearances, and squared distances from each segment to stationary centers. Half-side $\epsilon = \rho/10$ gives disjoint translation supports because $3\sqrt{2}\epsilon < \rho \leq \sqrt{\rho}$. In Euclidean space park on the horizontal axis to the right of every endpoint and of 2. For a transfer use a bend (m, m^2) , where the integer $m \geq 2$ exceeds the Cauchy root bounds of all polynomials obtained by restricting forbidden lines to that parabola. Each polynomial is nonzero of degree at most two, so this terminates. A rational $\epsilon < 1$, below every endpoint half-side and one tenth of the minimum squared segment clearance, gives disjoint supports. A branch joining nonhalt rectangles has centers at least 2 throughout.

In either routing, shrink linearly to the small squares, translate them in sequence, and expand linearly in the target rectangles, using the closed-rectangle cutoff implementation in Lemma 6.1. Put all slots inside $[1/4, 3/4]$. The normalized affine endpoint maps are exact, and the above clearances show that every nonhalt torus path has $x < 1/2$, while every nonhalt Euclidean path has $x > 0$. The empty list uses zero.

Let E_0, y_* be the rational initial planar codes. On the torus load from $a = (1/16, 1/16)$ by the spatially constant field $\eta(t)(E_0 - a)$, with η flat from zero to one in $[1/4, 3/4]$. Follow by the repeated rule pulse and lift with $h(z) = \sin(2\pi z)/(2\pi)$. In Euclidean space load the origin along

$\theta(t)y_*$ with a compact moving cutoff equal to one on a square of half-side one and supported in its enlargement by one. Follow by the repeated planar pulse and lift with $h(z) = z\chi(z)$, where $\chi = 1$ on $[-1, 1]$ and is supported in $(-2, 2)$. Lemma 6.2 gives the exact planar motions and Lemma 3.4 gives the forces. Its bounded-velocity comparison refinement is exactly the second row's uniqueness class.

At time one both particles are their initial codes. Subsequent rule samples encode the run. In the torus case the n th compute visit is at $1 + \sum_{j < n} (2(T_j - h_j) + 1)$ with $T_j = j + 2$; the Euclidean case has $T_j = j + 1$. A halt code satisfies the observer, including an initial halt after loading. In a nonhalting run the torus loader stays below $1/4$, and the Euclidean loader has nonnegative first coordinate; all later paths obey the strict bounds proved above. The history block's known absolute starting cell a recovers absolute head location as well as the head-relative work tape. All support and derivative bounds come from finitely many prescribed pulses. $\square$

9 Marked tape words and a separate history stack

The tape-track recorders retain the indexed work tape while a head travels to the history frontier. Here the entire finite work interval is stored in the first stack and history in the second. To reach the marked head, the processor transfers the preceding work prefix onto the history stack, performs the local update, and restores that prefix. Its geometric realization separates horizontal scaling into two successive operations, making the intermediate observer bound explicit. Normalize to one halt state h and denote the work alphabet by Γ .

For each nonhalt instruction $(q, a) \mapsto (q', b, d)$ make one case i if $d = 0$, and, for each of $d = \pm 1$, one case for each neighbor letter $c \in \Gamma$ and one end case. Write $q^+(i) = q'$ for the destination state of case i . Use alphabet

$$\mathcal{A} = \Gamma \sqcup \{\bar{a} : a \in \Gamma\} \sqcup \{E, \#\} \sqcup \{J_i\}_i \sqcup \{J_*\},$$

where bars mark the head and E pads the finite tape word. The controls are P_q, S_q (the latter only for nonhalt q), and R_i .

At a P_q visit the first stack is a finite tape interval in its natural left-to-right order, with exactly one marked head, followed by E^∞ . The second is finite history followed by E^∞ . Initially the interval is cells $0, \dots, \max(0, |w| - 1)$, with its first symbol marked; empty input gives the one-letter word $\bar{\square}$. The initial control is P_{q_0} and the second stack is E^∞ . The S_q scan transfers the word left of the mark, the update writes and moves the head mark (adjoining a blank precisely at an endpoint), and the R_i scan restores the left part. Replacing $\#$ by J_i finishes the step. If the left-of-head lengths before and after are L_-, L_+ , its count is $3 + L_- + L_+$. History determines the absolute left endpoint as minus the number of left-end extensions. At P_h the halt loop merely adds J_* forever.

Entry and scan rules are

$$(P_q; \epsilon, \epsilon) \rightarrow (S_q; \epsilon, \#) \quad (q \neq h), \quad (P_h; \epsilon, \epsilon) \rightarrow (P_h; \epsilon, J_*),$$

$$(S_q; c, \epsilon) \rightarrow (S_q; \epsilon, c) \quad (c \in \Gamma).$$

For the cases of $(q, a) \mapsto (q', b, d)$ the updates are

$$\begin{aligned} d = 0 : \quad & (S_q; \bar{a}, \epsilon) \rightarrow (R_i; \bar{b}, \epsilon), \\ d = 1 : \quad & (S_q; \bar{a}c, \epsilon) \rightarrow (R_i; \bar{c}, b), \quad c \in \Gamma, \\ & (S_q; \bar{a}E, \epsilon) \rightarrow (R_i; \bar{\square}E, b), \\ d = -1 : \quad & (S_q; \bar{a}, c) \rightarrow (R_i; \bar{c}b, \epsilon), \quad c \in \Gamma, \\ & (S_q; \bar{a}, \#) \rightarrow (R_i; \bar{\square}b, \#). \end{aligned}$$

Complete them by

$$(R_i; \epsilon, c) \rightarrow (R_i; c, \epsilon) \quad (c \in \Gamma), \quad (R_i; \epsilon, \#) \rightarrow (P_{q^+(i)}; \epsilon, J_i).$$

All tested and produced prefixes have length at most two.

For the full-domain check, ordinary versus marked first-stack tops separate scans and updates at S_q ; the displayed neighbor tests separate the update cases. Right tops separate the R_i rules. Into S_q , entry writes $\#$ on the right while scans write ordinary letters. Into R_i , its unique update has a marked first-stack top and return loops have ordinary first tops. Into P_q , distinct J_i distinguish case exits and J_* distinguishes the halt loop. Thus full source and image cylinders are separately disjoint, and the initialized machine simulation is faithful indefinitely.

Theorem 9.1 (Marked words and a fixed middle arc). *This recorder gives an effective smooth mean-zero general force on the unit torus, with all mixed derivatives bounded and period one for $t \geq 1$. Its unique classical periodic zero-data solution is global with bounded energy. The particle from $(1/4, 1/4, 0)$ enters $(1/2, 3/4) \times \mathbb{T}^2$ exactly when the machine halts. The repeated protocol uses only the table and initialization uses the input. Separately, one may choose either a solenoidal force with the same eventual periodicity, or a mean-zero force satisfying $\|f(t)\|_{C^m} = O((1+t)^{-1})$ for every spatial order m , with all mixed derivatives bounded and hence temporal L^2 in every C^m norm. These two alternatives are separate prescriptions.*

Proof. Use odd digits and base $K = 2|\mathcal{A}| + 1$. If there are N controls, let $\sigma_0 = 1/[100(N+1)]$, number them by $j(s)$, and put

$$o_s = (x_s, 1/4 + 3\sigma_0 j(s)), \quad x_s = \begin{cases} 5/8, & s = P_h, \\ 1/4, & s \neq P_h. \end{cases}$$

Encode by $(o_s + \sigma_0(E(A), E(B)), 0)$. State squares are separated vertically by $2\sigma_0$. Within a state, Lemma 3.1 gives a source or target coordinate gap at least $\sigma_0 K^{-2}$. Write D_r, D'_r for the matched rectangles, c_r, c'_r for their centers, and

$$\lambda_r = K^{|\alpha| - |\alpha'|}, \quad \mu_r = K^{|\beta| - |\beta'|}, \quad F_r(y) = c'_r + \text{diag}(\lambda_r, \mu_r)(y - c_r).$$

All side lengths are at most σ_0 .

We describe five divergence-free fields whose successive unit-mass pulses realize F_r . Let ρ_D equal one on the $\delta/2$ enlargement of D and be supported in its δ enlargement, where $\delta = \sigma_0/(10K^2)$. Put $A_D = \partial_x((x - c_1)\rho_D)$; it is one on D and has zero integral. For M rules, numbered $j_r = 0, \dots, M-1$, assign

$$z_r = 1/4 + \frac{j_r + 1}{2(M+1)}, \quad m_z = \frac{1}{16(M+1)}.$$

Let $\theta_r(z)$ be one on $[z_r - m_z, z_r + m_z]$, supported in its m_z enlargement, and set $B_r = \partial_z((z - z_r)\theta_r)$. The layer supports are disjoint, $B_r = 1$ near z_r , and $\int B_r = 0$. Take η_r to be the product of θ_r and cutoffs equal to one on $[c_{ra} - \sigma_0, c_{ra} + \sigma_0]$ with margin σ_0 , for $a = 1, 2$. Define

$$H_r^x = (\log \lambda_r)(x - c_{r1})(z - z_r)\eta_r, \quad H_r^y = (\log \mu_r)(y - c_{r2})(z - z_r)\eta_r,$$

$$\begin{aligned}
W_1 &= (0, 0, \sum_r z_r A_{D_r}), \\
W_2 &= \sum_r (\partial_z H_r^x, 0, -\partial_x H_r^x), \\
W_3 &= \sum_r (0, \partial_z H_r^y, -\partial_y H_r^y), \\
W_4 &= \sum_r B_r(z)(c'_r - c_r, 0), \\
W_5 &= (0, 0, -\sum_r z_r A_{D'_r}).
\end{aligned}$$

All functions are smooth torus functions by zero extension of their compact chart factors. Every field is divergence free: mixed partials cancel in W_2, W_3 , and the others are constant in their flow directions. Their means vanish by the derivative formulas, including $\int B_r = 0$. For $1 \leq j \leq 5$ let β_j be the derivative of a flat step from zero to one on $[(2j-1)/12, 2j/12]$, and put $W(\tau) = \sum_j \beta_j(\tau) W_j$, periodically extended. These disjoint pulses are nonnegative and have integral one.

Starting from $(y, 0)$, $y \in D_r$, the first pulse raises it to (y, z_r) . On that layer the second field is

$$((\log \lambda_r)(x - c_{r1}), 0, -(\log \lambda_r)(z - z_r)),$$

so it scales the first offset by λ_r and keeps $z = z_r$. At intermediate pulse mass $b \in [0, 1]$ its factor is λ_r^b ; the offset is at most $\sigma_0/2$ because both endpoint widths are at most σ_0 . Thus the candidate path stays on the asserted plateau. The third pulse analogously scales the second offset by μ_r , still on its plateau. The fourth translates by $c'_r - c_r$, since $B_r(z_r) = 1$ and other layers vanish. Target separation makes the fifth pulse lower the point back to zero. Smooth ODE uniqueness therefore gives $(y, 0) \mapsto (F_r(y), 0)$ on every full source rectangle. For a branch whose two controls are nonhalting, the first coordinate throughout lies in

$$[1/4 - \sigma_0/2, 1/4 + 3\sigma_0/2],$$

both while scaling and on the subsequent translation. It avoids the observed arc. On the halt loop $\lambda_r = 1$ and $c'_{r1} = c_{r1}$, so the first coordinate stays unchanged throughout.

Let y_{in} be the rational initial code. During $[0, 1]$ use $\beta_0(t) \cos(2\pi z)(y_{\text{in}} - (1/4, 1/4), 0)$, where β_0 is a unit-mass flat pulse on $[1/4, 3/4]$. It is divergence free and mean zero, and the distinguished particle travels on $z = 0$ to its initial code. Follow with $W(t-1)$. The resulting velocity V is smooth at the joins, starts from zero, and has bounded derivatives. Lemma 3.4 gives its residual force and pressure zero. The exact rectangle maps give the rule-code induction at times $1+k$. A halt lies in $[5/8, 5/8 + \sigma_0] \subset (1/2, 3/4)$; a nonhalting run has a safe loader and stays in the displayed band at all subsequent times. This proves the first force contract.

For the decaying choice use the onto clock $s = \log(1+t)$ and $\hat{V}(t) = aV(s)$, $a = (1+t)^{-1}$. Its force is

$$\hat{f} = a^2(V_s - V + (V \cdot \nabla)V)(s) - \nu a \Delta V(s).$$

Bounded derivatives of V give the asserted C^m estimate; every time derivative is a finite sum of bounded derivatives of V times powers of a , so mixed derivatives are bounded. Integrating $(1+t)^{-2}$ gives temporal L^2 . Since s covers all nonnegative internal time, the same exact material event is retained. This choice replaces the periodic forcing; it does not impose decay on that periodic field. For the separate solenoidal periodic choice, let $\Delta\phi = \text{div}((V \cdot \nabla)V)$, $\int \phi = 0$. Proposition 2.4 gives $f_{\text{sol}} = \partial_t V + (V \cdot \nabla)V - \nabla\phi - \nu\Delta V$ and $p = -\phi$, preserving all periodic bounds and the velocity. All cutoffs, logarithms of positive rational scale factors, and finite differentiations are effective; none depend on a future executed branch. $\square$

10 History placement when a prefix changes area

The position of a history record determines which prefixes an instruction replaces and when its scans terminate. If it replaces prefixes of lengths a, b by prefixes of lengths a', b' , its planar determinant contains $B^{a+b-a'-b'}$, where B is the radix. The following processors retain these length changes explicitly and use the geometric tools already proved. Our goal is the family of fixed-label halting events stated in Theorem 10.2 below. We first prove the symbolic instructions, then realize their full branch domains in Section 11, and finally verify initialization and continuous-time observation in Section 12.

Two geometric conclusions will be useful. A compensating third scale realizes the instruction on a three-dimensional neighborhood. Alternatively, a planar field can be lifted to an incompressible field with an invariant coding plane; only the motion in that plane is then prescribed. We will keep these conclusions separate. We will also distinguish a loader used once from a boot instruction that belongs to the repeated program.

The histories below implement different choices about where to store the erased instruction and when to scan to it. The main configurations and the stopping condition of each scan come before its finite table. The inverse checks concern every allowed configuration, including those outside an initialized computation. This is the information needed to turn the symbolic branches into disjoint closed rectangles.

Throughout this section a machine has a finite tape alphabet A , blank $\square$, finite state set Q , initial state q_0 , and halting states H . Its tape is indexed by $\mathbb{Z}$, its head starts at 0, and the finite input w is written from cell 0 rightwards. For $q \notin H$ its table is

$$\delta(q, a) = (q', b, d), \quad d \in \{-1, 0, 1\}.$$

A missing transition may be replaced by a transition to a halting state, writing back the scanned symbol and staying in place. Several halting states may be merged when convenient. These finite changes preserve halting, including an initially halted machine. An instruction

$$(s; \alpha, \beta) \mapsto (s'; \alpha', \beta')$$

acts on every pair of infinite tails: $(s, \alpha L, \beta R)$ is sent to $(s', \alpha' L, \beta' R)$. Empty prefixes are denoted by ϵ . Two prefixes are incompatible if they differ at a position specified by both. Our instruction tables have separately disjoint source cylinders and image cylinders, including the control state in each cylinder.

Remark 10.1 (A one-sided work tape). The two-sided tape convention also implements a one-sided machine whose left move at cell zero means staying at zero. Add a boundary bit to the work alphabet, set it to one at cell zero and zero everywhere else, and preserve it when writing. If a transition requests a left move while reading boundary bit one, replace that displacement by zero; otherwise retain its displacement. The head never visits a negative cell. Induction gives exactly the original one-sided work configuration after each transition, so all subsequent history compilers apply without changing halting. This preprocessing also leaves a prescribed stay instruction unchanged. The extra bit is a work symbol component, independent of each compiler's history track and return marker.

The theorem groups the constructions by their geometric domains. The row names distinguish the history placement, loading, and routing choices developed below; each row has its own fixed label and detector. The proof is deferred to Section 12, after the processor and geometric constructions.

Theorem 10.2 (Box, sheet, and invariant-plane realizations). *For every deterministic machine and finite input, and every fixed positive computable viscosity ν , each row of the following table has a finite effective prescribed force $f = \mathcal{F}_\nu[U]$. The initial velocity is zero. The force and velocity are smooth with all mixed derivatives bounded; the velocity has bounded kinetic energy. On $\mathbb{R}^3$ their spatial supports lie in one compact set independent of time, allowed to depend on the finite instance. On the unit torus both fields have spatial mean zero. Their global solution is $(u, p) = (U, 0)$. On the torus it is unique in the classical periodic class. For the Euclidean rows it is unique among smooth classical solutions with $u \in C_t H^2 \cap C_t^1 L^2$, $p \in C_t H^1$, and bounded $u, \nabla u$ on every finite time interval; pressure representatives differing by spatial constants have the same velocity conclusion. These are the specified cases of Lemma 2.2.*

The material trajectory from the listed fixed label a enters the listed fixed open set O if and only if the machine halts. In a torus row, the displayed interval is an arc of $\mathbb{R}/\mathbb{Z}$ and O is that interval times $\mathbb{T}^2$.

Construction	Domain and fixed label a	Observation O	Period one from
Euclidean full boxes			
Left-word boxes	$\mathbb{R}^3, (0, 0, 0)$	$x_1 < -1$	$t = 1$
Right-word boxes	$\mathbb{R}^3, (0, 0, 0)$	$x_1 < -1/2$	$t = 1$
Tagged storage	$\mathbb{R}^3, (-4, 0, 0)$	$(-1, 2)^2 \times (-1, 1)$	$t = 1$
Boot-window prisms	$\mathbb{R}^3, (-3, 0, 0)$	$x_1 > 0$	$t = 0$
Reserved-loader sheets			
Left reserved sheets	$\mathbb{T}^3, (1/2, 1/2, 1/4)$	$(1/8, 1/4)$	$t = 0$
Right reserved sheets	$\mathbb{T}^3, (1/2, 1/8, 1/4)$	$(3/4, 15/16)$	$t = 0$
Invariant planes			
Frontier sine plane	$\mathbb{T}^3, (1/2, 1/4, 0)$	$(5/8, 15/16)$	$t = 1$
Previous-tag plane	$\mathbb{T}^3, (1/8, 1/4, 0)$	$(2/3, 5/6)$	$t = 0$
Shifted-start plane	$\mathbb{T}^3, (3/4, 3/4, 1/2)$	$(0, 1/3)$	$t = 0$
Separator sine plane	$\mathbb{T}^3, (1/4, 1/2, 0)$	$(1/2, 1)$	$t = 1$
Two-cell-frontier plane	$\mathbb{T}^3, (1/8, 1/8, 0)$	$(2/3, 7/8)$	$t = 1$
Zero-blank plane	$\mathbb{T}^3, (1/4, 1/2, 1/2)$	$(3/4, 7/8)$	$t = 1$

For the rows beginning at $t = 1$, the repeated program depends only on the machine table, while the input appears in the loading pulse. The rows beginning at zero incorporate loading into the repeated prefix table, reserve a disjoint loader image, or place the initial code at the fixed label as specified in the proof. No force in this theorem is required to be solenoidal. The separator sine-plane row additionally permits an effective mean-zero solenoidal force with the same velocity and event. For arbitrary fixed real $\nu > 0$, the same algebraic formulas and conclusions hold, with numerical evaluation relative to that parameter.

10.1 Logging after an update

The first processor keeps a finite left tape word above a delimiter and hides the history below it. It handles a left move beyond that word by exposing an implicit blank.

Lemma 10.3 (Finite left word and a subsequent history sweep). *The delimiter-history table of Lemma 7.1, with $(N_q, D_i, U_q, \star)$ renamed $(P_q, T_i, U_q, \diamond)$, applies to any finite halt set H by omitting its normal rules at every $q \in H$. It simulates one machine transition in exactly $2|L'| + 3$ instructions, with L' the updated finite left word, and has separately disjoint full source and image cylinders. Starting with an empty left word, $|L_j| \leq j$ after j machine transitions.*

Proof. The displayed renaming is bijective and changes no write, displacement or top-letter guard in (11); it therefore conjugates the partial maps on all their cylinders. Allowing several halt states only omits their normal rules. The inverse distinctions at transfer, history-insertion and restoration

states are unchanged, so the full-domain proof of Lemma 7.1 applies to this extension. That lemma also gives the initialized simulation and the exact count. The finite left word increases in length by at most one per ordinary transition, hence $|L_j| \leq j$. Displacement records recover the absolute head position. It lies in $[-j, j]$ after j transitions, and every nonblank work cell lies in $[-j, |w| + j]$: each step writes at the old head and moves by at most one. These bounds recover the finite tape content in addition to the halting event. $\square$

The bottom history tail may be $\#^\infty$, $\square^\infty$, or a reserved letter $h_*\square^\infty$. None changes a rule: the history scan stops at the exposed delimiter. More precisely, on the invariant sets reached from these starts, replace the bottom tail while retaining the finite record list. This replacement commutes with every instruction. The following applications choose the tail according to their loading geometry; the symbolic inverse and count remain those just proved.

10.2 Logging the previous tag before an update

A second table incorporates loading into the repeated program and logs the previous tag before executing the next transition. This changes both the location of history and the exact instruction count.

Lemma 10.4 (Previous-tag processor). *There is an injective partial prefix processor starting from a distinguished state I with both stacks $\square^\infty$. One instruction loads the input. At a nonhalting main state with k stored left cells, the next machine transition takes exactly $2k + 4$ instructions.*

Proof. Give each update case of Lemma 10.3 its own tag λ , and add a reserved tag $\perp$. Use distinct letters h_λ , markers $\#, \diamond$, and states $B(q, \lambda)$, $U(q, \lambda)$, $V(q)$, $D(q)$, with the last three types needed only for nonhalting q .

At a main state the stacks are $L\#Z, R$. The first rule inserts $\diamond$, the $k = |L|$ transfers and k restores leave the tape unchanged, insert h_λ into Z , and remove $\diamond$. The final update is exactly the four-case tape update proved above. The instruction count is $1 + k + 1 + k + 1 + 1 = 2k + 4$. Write $\lambda_0 = \perp$ and let λ_j be the j th update tag for $j \geq 1$. After $j \geq 0$ transitions the current tag is λ_j and the history is $Z = h_{\lambda_{j-1}} \cdots h_{\lambda_0} \square^\infty$, with an empty recorded prefix when $j = 0$. Thus the loaded configuration has tag $\perp$ and blank history, while for $j \geq 1$ the recorded prefix ends at $h_\perp$. This recovers the displacement history and proves the configuration and halting correspondence, including an initially halted machine after the one loading instruction.

The complete rule list begins

$$\begin{aligned}
(I; \epsilon, \epsilon) &\mapsto (B(q_0, \perp); \#, w), \\
(B(q, \lambda); \epsilon, \epsilon) &\mapsto (U(q, \lambda); \epsilon, \diamond), \quad q \notin H, \\
(U(q, \lambda); c, \epsilon) &\mapsto (U(q, \lambda); \epsilon, c), \quad c \in A, \\
(U(q, \lambda); \#, \epsilon) &\mapsto (V(q); \#h_\lambda, \epsilon), \\
(V(q); \epsilon, c) &\mapsto (V(q); c, \epsilon), \quad c \in A, \\
(V(q); \epsilon, \diamond) &\mapsto (D(q); \epsilon, \epsilon).
\end{aligned}$$

For a transition $\delta(q, a) = (q', b, d)$ the rules from $D(q)$ are

$$\begin{array}{r|l|l}
1 & (D(q); \epsilon, a) & (B(q', \lambda); b, \epsilon) \\
0 & (D(q); \epsilon, a) & (B(q', \lambda); \epsilon, b) \\
-1, c \in A & (D(q); c, a) & (B(q', \lambda); \epsilon, cb) \\
-1, c = \# & (D(q); \#, a) & (B(q', \lambda); \#, \square b).
\end{array}$$

There are no outgoing rules from halting $B(q, \lambda)$. The tags make every incoming branch to a B -state unique, including the loader, whose tag is $\perp$. Images at U are separated by right tops $\diamond$ versus $c \in A$; images at V by left prefixes $\#h_\lambda$ versus $c \in A$; into $D(q)$ there is one rule. Source separation is given by the displayed top-letter tests. Consequently both cylinder families are disjoint. $\square$

10.3 A finite right word extended by a blank

The previous processors store only finitely many left cells. The following one instead keeps the left half-tape infinite and the right tape window finite. Its scan explicitly appends a blank before returning.

Lemma 10.5 (Right-word processor). *An effective injective prefix table represents a machine configuration by*

$$(q, L, r\#h\Box^\infty), \quad L \in A^\mathbb{N}, \quad r \in A^*, \quad |r| \geq 1,$$

where r starts at the head and all unlisted cells to its right are blank. If the local write and move leave a word v before $\#$, one simulated transition uses exactly $2|v| + 3$ instructions and replaces the right stack by $v\Box\#\tau_e h\Box^\infty$.

Proof. Index $e = (q, \ell, a)$ for every nonhalting q and $\ell, a \in A$, and write $(q_e, b_e, d_e) = \delta(q, a)$. Add $\#, \diamond, \tau_e$ and states F_e, J_q . The first rule has source $(q; \ell, a)$ and target

$$\begin{cases} (F_e; \diamond b_e \ell, \epsilon), & d_e = 1, \\ (F_e; \diamond, \ell b_e), & d_e = -1, \\ (F_e; \diamond \ell, b_e), & d_e = 0. \end{cases}$$

Add

$$\begin{aligned} (F_e; \epsilon, c) &\mapsto (F_e; c, \epsilon) && (c \in A), \\ (F_e; \epsilon, \#) &\mapsto (J_{q_e}; \epsilon, \Box\#\tau_e), \\ (J_q; c, d) &\mapsto (J_q; \epsilon, cd) && (c, d \in A), \\ (J_q; \diamond, \epsilon) &\mapsto (q; \epsilon, \epsilon). \end{aligned}$$

All source tests are disjoint. Into F_e the entry starts the left stack with $\diamond$, whereas each loop starts it with a distinct tape letter. Into J_q the right prefixes $\Box\#\tau_e$ are separated from one another by their tags and from every cd by the second letter; the cd themselves distinguish loop images. There is one incoming rule to q .

Writing $r = ar_1$, the word v is respectively r_1 , $\ell b_e r_1$, or $b_e r_1$. The left stack below the inserted $\diamond$ is exactly the updated infinite left tape. The F_e loop moves the finite word v onto that stack in reverse order. The delimiter rule adds a blank and records e . The J_{q_e} loop restores v while its second-stack first letter remains in A , and its exit removes $\diamond$. There are $1 + |v| + 1 + |v| + 1$ rules. In particular the possibly empty v after a right move causes no undefined rule: the inserted blank supplies the next scanned cell. Start with $L = \Box^\infty$, $r = w\Box$, and empty history. Induction proves the claimed tape interpretation and exact halting equivalence. History length gives the number of simulated transitions and their displacements. After k transitions the head lies in $[-k, k]$ and no nonblank tape cell lies outside $[-k, |w| + k]$. Reading the finite right word up to $\#$ and the first $2k$ letters of L therefore recovers all possibly nonblank tape cells: a left offset greater than $2k$ has absolute index less than $-k$. The record list supplies the absolute head index. $\square$

10.4 Tagged transfer without a temporary right delimiter

Lemma 10.6 (Tagged left-word sweep). *There is an injective prefix processor whose update sweeps the finite left word through a disjoint tagged alphabet. Its exact transition count is $2m + 3$, where m is the updated left-word length.*

Proof. Use main labels M_q , branch labels D_e , restore labels C_q , a delimiter $\#$, distinct records k_e , and a disjoint tagged copy $\bar{A} = \{\bar{a} : a \in A\}$.

At M_q the stacks are $L\#Z, R$, starting at $\#^\infty, w\Box^\infty$. Each main rule performs precisely the head-relative tape update; for a right move the lookahead c is retained and becomes current. The m updated left letters are transferred with tags, the record is inserted below $\#$, and the m tagged letters are restored and untagged. The untagged right top then triggers the exit. This gives $2m + 3$ instructions and the next main configuration. Every scan is finite, and no untagged tape data are confused with the tagged scan segment. The same induction as in Lemma 10.3 proves the halting equivalence and reconstruction of head displacement.

A right-moving transition is split by its next right letter:

$$(M_q; \epsilon, ac) \mapsto (D_e; b, c), \quad c \in A.$$

For a left move use $(M_q; \alpha, a) \mapsto (D_e; \epsilon, \alpha b)$ for $\alpha \in A$, and $(M_q; \#, a) \mapsto (D_e; \#, \Box b)$. For a stay use $(M_q; \epsilon, a) \mapsto (D_e; \epsilon, b)$. Every displayed case has its own e , with destination $q'(e)$. Complete the table by

$$\begin{aligned} (D_e; \alpha, \epsilon) &\mapsto (D_e; \epsilon, \bar{\alpha}), \quad \alpha \in A, \\ (D_e; \#, \epsilon) &\mapsto (C_{q'(e)}; \#k_e, \epsilon), \\ (C_q; \epsilon, \bar{\alpha}) &\mapsto (C_q; \alpha, \epsilon), \quad \alpha \in A, \\ (C_q; \epsilon, a) &\mapsto (M_q; \epsilon, a), \quad a \in A. \end{aligned}$$

Sources separate by their indicated letters and the right lookahead. Into D_e , the main-rule image has an untagged right top while loop images have distinct tagged tops. Into C_q , the left prefixes $\#k_e$ separate entries from loops with untagged left tops. The incoming rules to M_q preserve distinct right tops a . These facts prove full-cylinder injectivity. $\square$

10.5 A finite tape window and a boot instruction

This processor stores a finite tape window in one stack and its history in the other. Its fixed initial code has both stack coordinates equal to zero, which will make the entire velocity periodic from the initial time.

Lemma 10.7 (Window processor). *There is an effective injective prefix table with a boot state s_* and main states N_q . From $(s_*, E^\infty, E^\infty)$ it loads the input in one instruction. At a main configuration*

$$(N_q, \ell Jar E^\infty, \sigma g_* E^\infty)$$

it represents the finite tape window ℓar , with head at a ; both words ℓ, r are in left-to-right order and all cells outside the window are blank. One machine transition takes $3 + |\ell| + |\ell_{\text{new}}|$ instructions.

Proof. Use fresh markers J, E , a history base g_* , and records g_τ . A tag τ specifies (q, a, d, c) for a moving rule, where $c \in A$ is the neighboring tape letter or $c = \perp$ denotes a window edge; a stay

has tag $(q, a, 0, \perp)$. Use scan states S_q for nonhalting q and restore states D_τ . With $w_0 = w$ for nonempty w and $w_0 = \square$ otherwise, the boot and scan rules are

$$\begin{aligned} (s_*; \epsilon, \epsilon) &\mapsto (N_{q_0}; Jw_0E, g_*), \\ (N_q; \epsilon, h) &\mapsto (S_q; \epsilon, h), \quad q \notin H, \quad h \in \Lambda, \\ (S_q; c, \epsilon) &\mapsto (S_q; \epsilon, c), \quad c \in A, \end{aligned}$$

where $\Lambda = \{g_*\} \cup \{g_\tau\}$. For $\delta(q, a) = (q', b, d)$ the update rules are

$$\begin{array}{l|l|l} 0 & (S_q; Ja, \epsilon) & (D_{(q,a,0,\perp)}; Jb, \epsilon) \\ 1, c \in A & (S_q; Jac, \epsilon) & (D_{(q,a,1,c)}; Jc, b) \\ 1, \text{ edge} & (S_q; JaE, \epsilon) & (D_{(q,a,1,\perp)}; J\square E, b) \\ -1, c \in A & (S_q; Ja, c) & (D_{(q,a,-1,c)}; Jcb, \epsilon) \\ -1, \text{ edge} & (S_q; Ja, h) & (D_{(q,a,-1,\perp)}; J\square b, h), \quad h \in \Lambda. \end{array}$$

Finally include $(D_\tau; \epsilon, c) \mapsto (D_\tau; c, \epsilon)$ for $c \in A$ and $(D_\tau; \epsilon, h) \mapsto (N_{q'_\tau}; \epsilon, g_\tau h)$ for $h \in \Lambda$.

The domains separate by history tops at N , by tape tops versus J at S , by the tested current letter and neighbor, and by second-stack tops at D . Into S_q , history tops distinguish entry from tape-letter scan images. Into D_τ , first tops J distinguish updates from restore loops; the only multiple update entries sharing one tag are the left-edge variants, whose second tops h distinguish their images. Into a main state the prefixes $g_\tau h$ distinguish every restore image; the boot image instead starts with g_* . Hence both full cylinder families are disjoint.

Entering S_q and transferring ℓ leaves $JarE^\infty$ and puts $\text{rev}(\ell)$ above the old history. A stay replaces a by b . A right move pushes b onto this reversed left segment and takes the next right cell as current, inserting $\square$ if at the edge. A left move pops the nearest left cell, or uses a new blank if there is none, and puts b at the front of the right segment. The D_τ loops restore the new left segment in its original order, and the exit records the tag. Counting entry, the $|\ell|$ scan moves, the update, $|\ell_{\text{new}}|$ restore moves, and the exit proves the formula. This also proves the invariant and that every simulated step is finite. The tag list determines absolute head position. An initially halted machine is detected immediately after boot; otherwise halting and nonhalting follow by induction. $\square$

10.6 Two finite-window history programs

The next two processors use an ordinary tape with data, history, and marker tracks. They differ in how the right history frontier advances. Retaining this distinction records two finite local implementations of the same tape computation.

Lemma 10.8 (Frontier moved by a separate instruction). *There is a partial one-head machine with alphabet $\Sigma = A \times \{\mathbf{e}, \mathbf{F}, \lambda_r : r \in \mathcal{R}\} \times \{0, 1\}$, where $\mathcal{R} = (Q \setminus H) \times A$, and states B_q, N_q, L_q, A_r, R_r . With the history frontier initially at absolute cell 2, it simulates transition number $n + 1$ in*

$$2(2 + n - h_{n+1}) + 4$$

auxiliary steps, where h_{n+1} is the new machine head position. For every target state, all incoming rules have the same direction and the written combined symbol distinguishes their old state and scanned symbol.

Proof. This is the recorder of [9, Lemma 2.1] with initial frontier $r_0 = 2$, under the following bijections of control states and history letters:

$$(S_q, A_r, R_r, F_q, L_q) \mapsto (B_q, A_r, R_r, N_q, L_q), \quad (E, P, [r]) \mapsto (\mathbf{e}, \mathbf{F}, \lambda_r).$$

The work alphabet and marker bit are unchanged. In rows one, two, three, six, and seven the history symbol is required to differ from the frontier; row four reads the frontier, and row five reads an empty history cell. These are exactly the guards in [9, Lemma 2.1]. The writes and displacements also agree row by row. Thus these bijections conjugate the entire partial transition, not only its initialized orbit. Its full-domain inverse, destination-determined incoming directions, and distinguishing written symbols transfer.

The initialization sets every history cell to $\mathbf{e}$ except for $\mathbf{F}$ at absolute cell 2, and sets every marker bit to zero. The checkpoint invariant of [9, Lemma 2.1] gives records at $2, \dots, n+1$, frontier $2+n$, and original work head h_n . Its count $2(r_0 + n - h_{n+1}) + 4$ is the asserted count. The same invariant includes an initially halted machine. $\square$

To convert this table to two stacks, list cells to the left nearest first and cells from the head rightwards. For an auxiliary rule scanning g , writing g' , and moving d , use prefix pairs

d	source	target
1	(ϵ, g)	(g', ϵ)
0	(ϵ, g)	(ϵ, g')
-1	(e, g)	(ϵ, eg') , $e \in \Sigma$.

Determinism separates the sources. Within one target state the direction is fixed; the written symbol separates targets in the left coordinate for $d = 1$, in the right coordinate for $d = 0$, and in the two-letter right prefix (e, g') for $d = -1$. Thus this is again an injective full-cylinder table.

Lemma 10.9 (Two-cell frontier update). *There is an injective partial update on a centered tape with alphabet $\Sigma = A \times (\{\perp, \#\} \sqcup \mathcal{R}) \times \{0, 1\}$ and labels S_q, L_q, P_r, R_r . At the n th main checkpoint its frontier is $e = 2 + n$, and the next machine transition takes exactly $2(e - h_{n+1}) + 2$ updates. Each branch can be specified by the three source cells $-1, 0, 1$ and has a whole product prefix cylinder as its image.*

Proof. Write a tape cell as (a_j, g_j, m_j) with indices relative to the head. After the indicated edits a displacement d recenters by $s'_j = \tilde{s}_{j+d}$.

Initially the auxiliary head is at absolute cell 0 in S_{q_0} , all marker bits are zero, the history symbol is $\#$ at absolute cell 2 and $\perp$ everywhere else, and the work track is the input followed by blanks on the rest of the tape. At checkpoint n , the records occupy cells $2, \dots, n+1$, the frontier is at $e = 2 + n$, and the other history cells are empty. This holds at initialization. After the work update the new head $h' < e$ is marked, then the right scan finds e . Its two-cell rule records the transition and puts the frontier at $e + 1$, starting the return at $e - 1$. Along that return the cell to the right has index at most e , so it is not the new frontier; the loop guard is satisfied until the marked cell is reached. Clearing the marker gives the next checkpoint. The count is one work update, one marking update, $e - h' - 1$ outward loops, one frontier update, $e - h' - 1$ inward loops, and one exit, totaling $2(e - h') + 2$.

Use the following entire table:

label	condition	edits	d	new label
S_q	$a_0 = a$	$r = (q, a), a_0 \leftarrow b_r$	d_r	P_r
P_r	$m_0 = 0$	$m_0 \leftarrow 1$	1	R_r
R_r	$m_0 = 0, g_0 \neq \#$	none	1	R_r
R_r	$m_0 = 0, g_0 = \#, g_1 = \perp$	$g_0 \leftarrow r, g_1 \leftarrow \#$	-1	$L_{q'_r}$
L_q	$m_0 = 0, g_1 \neq \#$	none	-1	L_q
L_q	$m_0 = 1$	$m_0 \leftarrow 0$	0	S_q .

The first row is present only for nonhalting q .

Injectivity must hold beyond these intended runs. For output P_r , the record in the label determines the old state, overwritten work symbol, and displacement. Into R_r the output marker at index -1 distinguishes the marking entry from the loop. Into L_q , an incoming frontier update has a delimiter at output index 2 and a determining record at index 1; a continuation cannot have that delimiter because its old $g_1 \neq \#$. In the first case restore the old delimiter and empty following history cell; in the second simply undo the shift. Into S_q , restore a marker bit at index 0. These inverse rules prove injectivity of the entire partial update.

Fix the source symbols at $-1, 0, 1$ and the label wherever a row is defined. The edited source window, after shift d , occupies $-1 - d, \dots, 1 - d$, and its left and right prefix lengths are $1 + d$ and $2 - d$. Everything outside this window is an unrestricted unchanged tail, with the same order on its respective side. Thus the image is a whole prefix cylinder. Source cylinders partition the domain; injectivity makes their images disjoint. No assertion merely about one initialized orbit is used in passing to rectangles. $\square$

10.7 Exact codes and their changes of convention

Use the gapped coding of Lemma 3.1. For an alphabet Σ of size m , the notation in this section is

$$c(S) = E(S), \quad b(v) = E(v), \quad I_v = I(v).$$

Thus $c(vS) = b(v) + B^{-|v|}c(S)$, and eventually constant codes are rational. The lemma supplies injectivity and closed-prefix gaps for each digit convention listed below, including zero-digit boundary codes.

Choose a positive diagonal affine map E_s from $[0, 1]^2$ onto a state rectangle, with the state rectangles pairwise disjoint. The configuration code is $E_s(c(L), c(R))$. Every prefix rule prescribes the positive diagonal affine map

$$E_s(b(\alpha) + B^{-|\alpha|}u, b(\beta) + B^{-|\beta|}v) \mapsto E_{s'}(b(\alpha') + B^{-|\alpha'|}u, b(\beta') + B^{-|\beta'|}v). \quad (15)$$

This equality holds for every $(u, v) \in [0, 1]^2$. Separately disjoint symbolic cylinder families therefore give separately disjoint closed rectangle families, not merely separated initialized code points.

Lemma 10.10 (Exact equivalence of code conventions). *If two encodings use the same prefix table, possibly after a bijective renaming of letters and states, the map that sends each encoded state and pair of words to its other encoding conjugates the partial updates on their entire coded domains. On every branch, both extensions in (15) retain precisely the two tail coordinates. Changes of base, digits, or state rectangles require no identification of the surrounding fluid motions.*

Proof. The encodings are injective by the gap argument, so the indicated map and inverse are well defined on the coded sets. Applying one branch replaces exactly its two prefixes, leaving both words' tails unchanged in either convention; the two possible compositions therefore yield the same encoded output. Equation (15) proves the stronger full-rectangle identity in each encoding. This is a conjugacy of the specified partial symbolic dynamics; the separately given interpolation paths establish their respective all-time fluid events. $\square$

The encoding is also part of the loading argument. A zero blank digit can put the fixed initial point on a prefix boundary. Odd positive digits instead put eventually constant codes strictly inside their prefix intervals, leaving room for a reserved loading rectangle. The following choices record the two possibilities for the applications below.

Processor realization	Base	Digits
Left-word boxes	$2m + 2$	$1, 3, \dots, 2m - 1$
Reserved-loader sheets	$2m + 1$	$1, 3, \dots, 2m - 1$
Previous-tag plane	$2m + 1$	$1, 3, \dots, 2m - 1$
Frontier sine / shifted-start planes	$2m + 1$	$1, 3, \dots, 2m - 1$
Right-word boxes	$2m$	$0, 2, \dots, 2m - 2$; blank first
Tagged storage	$3m$	$1, 4, \dots, 3m - 2$
Separator sine plane	$2m + 1$	$1, 3, \dots, 2m - 1$
Boot-window prisms	$2m + 2$	$0, 2, \dots, 2m - 2$; E first
Two-cell-frontier plane	$2m + 2$	$1, 3, \dots, 2m - 1$
Zero-blank plane	$2m + 1$	$0, 2, \dots, 2m - 2$; blank first

Each row satisfies the proved digit hypotheses. Two reserved-loader sheet placements below use the same digit convention but different charts and fixed observers.

11 Normal compensation and continuous-time observation

The preceding tables give separate source and target rectangle families. They need not give a disjoint union of both families: a target can occupy another instruction's source region. The next construction avoids this conflict by separating instructions in height during the transfer. Its endpoint formula holds on a neighborhood of each complete box. This includes codes on rectangle boundaries, but it does not make that neighborhood invariant under later instructions.

For all finite-family constructions in this section, an empty instruction family means the zero repeated pulse. Empty pairwise minima are omitted; when a width or scale maximum has no entries we use the value one. An initially halted input is still carried to its halt code by the stated loading or boot operation. These conventions also cover singleton families without imposing an unnecessary pairwise test.

For two finite families of positive-width rational boxes $D_i^\pm$ centered in $z = 0$, write $c_i^\pm$ for their centers, $R_i^\pm$ for their horizontal projections, and $h_{ik}^\pm$ for their coordinate half-widths. Assume that within each family the horizontal projections are positively separated, as the prefix-cylinder checks above guarantee. Suppose $h_{ik}^+ = \lambda_{ik} h_{ik}^-$, all factors are positive, and $\prod_k \lambda_{ik} = 1$. These are exactly the hypotheses of Lemma 5.1, with source half-height h_{i3}^- and target half-height h_{i3}^+ . It permits the different thicknesses needed here; it prescribes the full diagonal map on a positive three-dimensional neighborhood.

For the additive-margin realization choose

$$H_3 = \max_{i,\pm} h_{i3}^\pm, \quad g = \min \left(\{1\} \cup \left\{ \max_{k=1,2} (|c_{ik}^\pm - c_{jk}^\pm| - h_{ik}^\pm - h_{jk}^\pm) : i < j \right\} \right).$$

The horizontal separation gives $g > 0$. Take $\eta = g/4$ and $Z_i = (2H_3 + 2)i$, using successive flat switches on $(1/8, 1/4)$, $(3/8, 1/2)$ and $(3/4, 7/8)$. Endpoint gaps lose only $2\eta \leq g/2$, while a height gap exceeds $2H_3 + 2\eta$. Thus these particular margins and heights satisfy every separation and plateau condition of that proof. The centers interpolate horizontally and the half-widths interpolate geometrically, so its observer bounds apply without any change of physical coordinates.

The normal scale has an explicit symbolic meaning. For an unscaled state interval $[o_s, o_s + 1]$ and a prefix rule, the horizontal factors are

$$\lambda_1 = B^{|\alpha| - |\alpha'|}, \quad \lambda_2 = B^{|\beta| - |\beta'|}, \quad \lambda_3 = (\lambda_1 \lambda_2)^{-1}.$$

Source thickness $[-1, 1]$ and target thickness $[-\lambda_3, \lambda_3]$ therefore satisfy Lemma 5.1. A code lies at height zero at every instruction boundary, so it lies in the next source box regardless of the previous target thickness. No invariant three-dimensional neighborhood of the code set is asserted.

The same proof permits the following two concrete choices of collars and stage times. These choices change the interpolating field, while preserving its proved endpoint map and observer estimate. For the odd-digit left-word code let m_* be the largest prefix length and take relative padding $\eta_* = B^{-m_*}/4$. Enlarge every moving box by a factor $1 + \eta_*$. Endpoint horizontal widths are at most 1, so a separating gap loses at most η_* and stays positive. The heights

$$Z_i = 4(1 + \eta_*) \max_j \{1, \lambda_{j3}\} i$$

separate the middle-stage vertical extents. Use three successive ramps $r_j(t) = r(3t - j + 1)$, where $r(v) = \rho(v - 1/4)/(\rho(v - 1/4) + \rho(3/4 - v))$ and ρ is the flat function in (2). These are exactly the three lift, change, and lower stages of the proof, with geometric shape interpolation. For the boot-window code use instead $\gamma = B^{-m_*}$, additive padding $\gamma/4$, and private heights $4(R + 1)i$, $R = \max(1, \max_i \lambda_{i3})$, on three equal stages with progress $\sigma(3v - 1)$. Both choices meet the same disjoint-neighborhood proof, including every closed-box boundary point.

Lemma 11.1 (Storage and delivery for a bounded observer). *Suppose the horizontal source and target rectangles for a prefix table lie in state bands $[o_s, o_s + 1] \times [0, 1]$, where the unique halting band has $o_h = 0$ and every other band has $o_s \geq 3$. Give source boxes half-height*

$$h = \frac{1}{1 + \sum_j \lambda_{j3}}$$

and target boxes half-height $h\lambda_{j3}$. An effective divergence-free pulse implements their determinant-one affine maps in $7N$ slots. Every trajectory from a box whose source and target bands are nonhalting avoids $\mathcal{O} = (-1, 2) \times (-1, 2) \times (-1, 1)$ throughout the pulse.

Proof. If $N = 0$, use the zero pulse. Assume henceforth that $N > 0$. Every source vertical half-width is $h \leq 1$, and every target half-width is $h\lambda_{j3} < 1$; horizontal half-widths are at most $1/2$. Thus every endpoint half-width is at most one. Put storage centers $m_j = (-10 - 5j, 0, 0)$, horizontal storage squares of half-side one at these centers, and travel height $Z = 6$. The source projections together with these storage squares form a disjoint family, and so do the target projections together with the storage squares. Choose a rational margin $0 < \eta < 1/2$ so their respective enlargements remain disjoint.

Use $3N$ consecutive slots to take each source box in turn vertically up by 6, horizontally to its storage center at height 6, and down. A localized translation is the curl of

$$\frac{1}{2}\chi(x - c(t)) [c'(t) \times (x - c(t))],$$

where $\chi = 1$ on a positive neighborhood of the moving box and is supported in its η enlargement. For a half-width vector r , write $\chi_{r,\eta}$ for the product cutoff equal to one on the box of half-widths $r + \eta/2$ and supported in the box of half-widths $r + \eta$. Use another N slots to apply each diagonal scaling at its storage center, by the curl of

$$\frac{1}{3}\chi_{(1,1,1),\eta}(x - m_j)[\theta'(t)G_j(x - m_j) \times (x - m_j)], \quad G_j = \text{diag}(\log \lambda_{j1}, \log \lambda_{j2}, \log \lambda_{j3}).$$

Here $\text{tr } G_j = 0$ and θ is a flat progression from 0 to 1 on that slot. The exact scaling path is $m_j + \exp(\theta G_j)(x - m_j)$; all its half-widths stay at most one, inside the cutoff plateau. Finally use $3N$ slots to deliver the scaled boxes, one at a time, to their target centers by the reverse type of high route.

During each vertical leg the projected support is separated from every stationary box by the relevant source/storage or storage/target family. During a horizontal leg the support is above

$6 - 1 - \eta$, while stationary boxes lie in $[-1, 1]$ in height. Scaling supports are disjoint at the storage centers. Hence the other tracked boxes remain stationary at every slot; induction proves the precise composition on each complete source box and on a positive neighborhood, by the curl-motion lemma and the finite positive margins. Source-target intersections cause no conflict because all sources are stored before any target is occupied.

A nonhalting tracked point has first coordinate at least 3 in source and target footprints, and less than -1 in storage footprints. Its vertical legs and scaling stay over these footprints. Horizontal legs have height at least 5. Each possibility excludes $\mathcal{O}$, proving all-time avoidance. $\square$

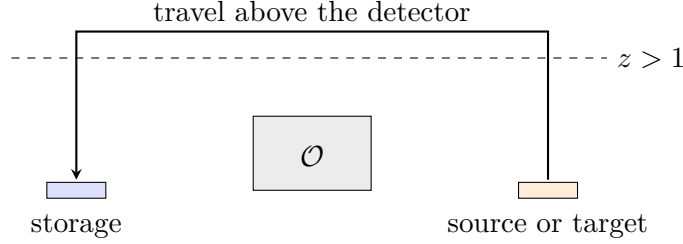


Figure 1: Storage routing for a bounded detector, with the second horizontal coordinate suppressed. All sources are evacuated before any target is occupied. Scaling takes place at the storage positions; horizontal travel takes place at height six. The drawing is schematic: the storage and endpoint footprints are separated by the explicit margins in Lemma 11.1.

11.1 Sheet and planar alternatives

When only a planar sheet carries the code, its vertical thickness can be chosen after the horizontal motion. This allows a stronger path statement: every horizontal coordinate of a tracked point can be made a convex combination of its own endpoints. The widths in the two horizontal directions need neither agree nor have equal products. For source and target horizontal centers a_i, b_i and half-widths r_{ik}^0, r_{ik}^1 , interpolate the center and both half-widths by the same parameter s :

$$c_{i,\text{hor}} = (1 - s)a_i + sb_i, \quad r_{ik} = (1 - s)r_{ik}^0 + sr_{ik}^1.$$

Take starting height $z_0 = 1/4$ and private heights $z_i = 1/2 + i/(4(N + 1))$. Lift, interpolate, and lower on disjoint time stages. A point of fixed fractional horizontal coordinates follows the convex combination of its own two endpoints. The affine generator is

$$c'_i(t) + \text{diag} \left(\frac{r'_{i1}}{r_{i1}}, \frac{r'_{i2}}{r_{i2}}, -\frac{r'_{i1}}{r_{i1}} - \frac{r'_{i2}}{r_{i2}} \right) (x - c_i(t)).$$

This is exactly the reciprocal-normal-strain construction in Lemma 3.3, applied inside the open unit cube to the stated rectangles. To verify its chart hypothesis for rectangles elsewhere than $[1/4, 3/4]^2$, take one tenth of the minimum of 1, the actual endpoint chart margins, the horizontal sup-norm gaps in the two endpoint families, and the private-height gaps. The expanded plates then remain inside the cube and are separated by footprints during lift and lowering, and by height during interpolation. The proof of that lemma consequently applies without a coordinate or viscosity normalization.

An invariant-plane realization solves a different problem. We first allow the planar field to compress the rectangles to small squares, route the squares through parking positions, and expand

at the targets. The lift will cancel its divergence in the normal direction. For planar routing we use Lemma 6.1 with $D = (0, 1)^2$ and the safe rectangle $G = (0, 1/2) \times (0, 1)$, or the right half of D as specified below. There are several explicit choices of the same finite path construction. They preserve the full-rectangle affine maps but have their own smooth paths:

1. For the frontier sine plane, use initial contraction on $[0, 1/4]$, $4N$ successive segment intervals on $[1/4, 3/4]$, and expansion on $[3/4, 1]$. The contraction factors are $1 - s + s\epsilon/r_{ik}$, and the expansion factors $1 - s + sr'_{ik}/\epsilon$. To avoid l forbidden lines, select a vertex among $2l + 3$ rational points $(a, 1/4 + a^2)$, $1/8 < a < 1/4$, also avoiding the two endpoints. A line contains at most two such points. Positive segment clearances give ϵ with moving and stationary squares separated by more than 10ϵ .
2. For the previous-tag plane, use $4N + 2$ equal stages and logarithmic contraction and expansion. Parking points have height half the minimum of all source and target center heights and abscissas $i/(2(N + 1))$. A two-segment vertex can be chosen below the finitely many forbidden rational lines: choose an abscissa below all positive vertical forbidden abscissas, then an ordinate below all positive intersections at that abscissa. Include the line through the two endpoints among the exclusions. With ϵ one hundredth of the minimum of 1, side lengths, boundary margins, and squared line distances, $100\epsilon \leq \min(1, d^2) \leq d$. A radial translation cutoff equal to one out to radius 2ϵ and zero past 3ϵ therefore misses every other tiny square of radius at most $\sqrt{2}\epsilon$.
3. For the shifted-start plane, choose all intermediate vertices in the right half. A horizontal line below every endpoint and obstacle cannot coincide with any forbidden endpoint-obstacle line; avoiding their finitely many intersections gives a rational vertex. Logarithmic contraction and expansion, with tiny half-side $\epsilon \leq 1/100$ and clearances greater than 10ϵ , keep a nonhalting point at first coordinate greater than $1/2 - \epsilon > 1/3$.
4. For the separator sine plane, use moving centers c_i and half-size matrices M_i throughout all stages, with linear interpolation during shrink and expansion. Relative padded boxes $c_i + \lambda M_i[-1, 1]^2$ are disjoint, and the field is

$$\sum_i \psi(M_i^{-1}(X - c_i))[c'_i + M'_i M_i^{-1}(X - c_i)].$$

Here $\lambda = 1 + d_0/4$ with d_0 a positive endpoint margin/gap bound. If d_1 bounds required boundary margins and $d_2 \leq 1$ bounds the positive squared center separations along all planned segments, choose $\epsilon = \frac{1}{2} \min(m_0, d_1/\lambda, d_2/(4\lambda))$, with m_0 the minimum half-side. Then padded tiny squares do not intersect, since their intersection would force center distance at most $2\sqrt{2}\lambda\epsilon < d_2$. The path $c_i + M_i v$ is exact for every $v \in [-1, 1]^2$.

5. For the two-cell-frontier plane, choose intermediate vertices on $(1/8 + r/8, 1/8 + r^2/8)$, $0 < r < 1$. If there are l forbidden lines, including the vertical lines through both endpoints, the $2l + 1$ samples $r = j/(2l + 2)$ include an admissible vertex. A line intersects the parabola in at most two points. Linear contraction and expansion and the one-hundredth clearance choice again prove full-rectangle transport and safe-region confinement.
6. For the zero-blank plane, choose a rational abscissa avoiding endpoint abscissas and vertical forbidden lines, then a rational ordinate avoiding the remaining line intersections. Use fixed source/target cutoffs, linear contraction and expansion, and tiny half-side $\epsilon = d_*/10$, where d_* is the minimum of 1, half-sides, boundary margins and positive squared segment-to-obstacle

distances. Actual distances are at least d_* , while support and stationary-square radii sum to $3\sqrt{2}\epsilon < d_*$. This includes boundary code points.

In every case the first round moves all source centers to distinct parking positions before the second round occupies targets. The cutoff equals one on a positive neighborhood of each tracked rectangle, or can be chosen with an arbitrarily smaller positive plateau collar without changing the field on that rectangle. The exact formulas for the finite affine stages then apply on a positive initial neighborhood. The stage fields vanish near stage endpoints, so they concatenate smoothly. These verifications explain precisely why the planar-routing lemma applies to all the listed variants; no abstract extension of an unspecified code map is being assumed.

For the trigonometric variants put

$$U(X, z, t) = \left(\cos(2\pi z)b(X, t), -\frac{\sin(2\pi z)}{2\pi} \operatorname{div}_X b(X, t) \right).$$

For the compact vertical variants use instead

$$U = (h'(z)b, -h(z) \operatorname{div}_X b), \quad h(z) = (z - 1/2)\chi(z),$$

where $\chi = 1$ on $[3/8, 5/8]$ and is supported in $[1/4, 3/4]$, periodically extended. Lemma 6.2 proves divergence cancellation, mean zero, and the exact planar motion on $z = 0$ or $z = 1/2$, respectively. The same formula applies to a spatially constant planar loading field: its divergence is zero, and the horizontal mean vanishes after multiplication by h' . Thus such loading does not need an unsupported compact planar chart assumption.

12 Initialization and fixed observation regions

A correct time-one instruction map does not yet prove a halting event. We must locate the initial code, make the particle reach it from a fixed label, and exclude the detector during every intermediate motion of a nonhalting computation. We now prove Theorem 10.2 by giving the actual placement and checking the event for each of its rows.

Proof of Theorem 10.2. We specify the geometric data and check the all-time events in three groups. Whenever one halting patch is specified, first merge the halting states as allowed above; this preserves the event, including initial halting. The positive digit gaps remain valid at every closed rectangle edge. In every case equation (15) makes the time-one map equal to the complete symbolic rule on its entire prefix rectangle. For boxes and moving sheets, the stated affine formula holds on a three-dimensional neighborhood. For an invariant-plane construction, planar routing gives the formula on a planar neighborhood, and the lift reproduces that motion exactly on its invariant plane; no prescribed affine map off that plane is inferred. Smooth flat collars allow the chosen pulse to be repeated every unit interval while preserving $U(0) = 0$.

Euclidean full boxes. For left-word boxes use Lemma 10.3 with initial stacks $\#^\infty, w\Box^\infty$, the odd-digit base $2m + 2$, and state offsets o_s given by distinct positive multiples of 3 for nonhalting labels and distinct negative multiples for halting main labels. The spatial code is $(o_s + c(L), c(R), 0)$. Use the relative-padding private-height construction following Lemma 5.1, and load the rational initial code from the origin by a localized straight translation during $[0, 1]$. A cutoff with a unit-box plateau and a larger compact support suffices. At a halt the first coordinate is at most -2 . On every nonhalting instruction both endpoint centers have first coordinate at least 3, while the first half-width is at most $1/2$ throughout the period. The entire path is therefore positive in that coordinate. A nonhalting loading segment is nonnegative. This proves the first row.

For right-word boxes use Lemma 10.5, even digits in base $2m$ with blank digit zero, offsets $-2j$ for the j th halting state and $2j$ for the j th other state, both numbered from 1. The initial code is $(o_{q_0}, b(w\Box\#), 0)$. The full-box data have unit source half-height and reciprocal target half-height, and use exactly the additive-gap construction of Lemma 5.1. Initialize by the same lemma for one translated unit cube with all scales equal to one. Its initial and final centers have height zero, although the loading path rises and descends; its first coordinate still interpolates from zero to the initial code. Halting gives first coordinate at most -1 . Nonhalting rule centers are at least 2 and half-widths at most $1/2$, excluding $x_1 < -1/2$ at all times.

For tagged storage use Lemma 10.6, digits $1, 4, \dots, 3m-2$ in base $3m$, and offsets $o_s = 3 \text{ index}(s)$ with the halting main state indexed zero. Use the thin boxes and $7N$ -slot storage pulse of Lemma 11.1. Load from $(-4, 0, 0)$ by three localized translations through

$$(-4, 0, 0), \quad (-4, 0, 6), \quad a_{\text{in}} + (0, 0, 6), \quad a_{\text{in}}.$$

Take unit-box plateaus and margin $1/4$ for these loading pulses. A nonhalting initial code has first coordinate at least 3, so the vertical loading legs are outside the observation box and the horizontal leg is at height 6. The storage lemma excludes it on every later nonhalting period. At a halting code both horizontal coordinates belong to $(0, 1)$ and the height is zero, so the code lies strictly inside the observation box.

For boot-window prisms use Lemma 10.7, base $2m+2$ with even digits and $d(E) = 0$, offset 2 for the halting main state, and distinct negative offsets $-3k$ for all others, with boot offset -3 . The initial code is exactly $(-3, 0, 0)$. Include the boot rule in the full-box pulse, using the private heights $4(R+1)i$ and padding specified above. Thus there is no loading interval outside the period. Halting codes have first coordinate in $[2, 3]$. Every nonhalting endpoint footprint lies at first coordinate at most -2 ; the center interpolation and half-width bound keep the whole moving box strictly negative. This proves the four Euclidean rows, including the initial-halt cases at time 1 after loading or boot.

For tagged storage, uniqueness also holds under the pointwise comparison bounds

$$u, \partial_t u, \nabla u, \nabla^2 u, p, \nabla p = O((1 + |x|)^{-2})$$

uniformly on every finite time interval, with pressure decaying at infinity. Smoothness and these bounds imply the Sobolev and time-continuity requirements of Section 2 by dominated convergence; $\partial_t u \in C_t L^2$ gives $C_t^1 L^2$. Hence the stated uniqueness includes this entire comparison class. For the other Euclidean rows, a pressure representative in $C_t H^1$, modulo spatial constants when applicable, is the normalization specified in the theorem. The constructed compactly supported velocity and zero pressure satisfy all these conventions.

Reserved-loader sheets. Both sheet rows use the post-update processor of Lemma 10.3, with an additional history symbol h_* unused in any ordinary rule. For the left reserved sheets, place the halting interval at $[5/32, 7/32]$ and enumerate the J other states by $j = 0, \dots, J-1$ with intervals

$$[1/2 + (3j+1)/(12J), 1/2 + (3j+2)/(12J)].$$

Use second interval $[1/4, 3/4]$ and height $1/4$. The input target is the code in U_{q_0} of $\#h_*\#^\infty, \diamond w\Box^\infty$. It lies in the interior of the unused image cylinder with prefixes $(\#h_*, \diamond)$. Indeed the incoming U_{q_0} images have left prefix $c \in A$ or $\#h_i$, never $\#h_*$. A rational square centered at this target, with half-side half its minimum coordinate distance to the four cylinder edges, is therefore disjoint from every ordinary target. Add a source square centered at $(1/2, 1/2)$ with half-side half the minimum of $1/4$ and its positive distances in the first coordinate to the state intervals. This source is disjoint from every ordinary source. The positive affine source-to-target map supplies a loader in the same repeated pulse.

For the right reserved sheets, use the same table with history tail $h_*\square^\infty$, halting interval $[13/16, 7/8]$, second interval $[1/4, 1/2]$, and height $1/4$. If S is the number of other labels, put $\delta = 1/(8(S+1))$ and give label $l = 1, \dots, S$ the first interval

$$[1/8 + 2(l-1)\delta, 1/8 + (2l-1)\delta].$$

These lie strictly below $3/8$. The loader source is the square at $(1/2, 1/8)$ with half-side $1/32$. Its target is the code of $(U_{q_0}, \#h_*\square^\infty, \diamond w\square^\infty)$, surrounded by the same half-minimum-margin square inside its reserved cylinder. The same prefix argument proves separate disjointness of both enlarged instruction families.

Apply the moving-sheet construction to each placement. The reciprocal normal strain is trace-free, all supports lie in the unit cube, and each horizontal coordinate of a tracked sheet point is a convex combination of its own endpoints. At integer time 1 the fixed particle is at its reserved U -code, and at time 2 the ordinary marker-removal instruction gives the initial main configuration. A machine halting initially is therefore detected at 2. Otherwise the main times satisfy $t_0 = 2$, $t_{j+1} = t_j + 2|L_{j+1}| + 3$. For the left placement, loading and all nonhalting endpoints lie at first coordinate at least $1/2$, and subsequent state intervals are in $(1/2, 3/4)$. For the right placement the loading point has coordinate $1/2$ and all nonhalting state intervals are below $3/8$. Convex interpolation excludes the corresponding observation arcs throughout every period. Halting state intervals are strictly inside the displayed arcs. Periodicity starts at zero because the same loader is present in every pulse, with no image collision.

Invariant-plane realizations. For the frontier sine plane use Lemma 10.8, its two-stack conversion, and odd digits in base $2m+1$. With P the auxiliary label set, put $s = 1/(8|P|)$ and choose an index $i(p) \in \{0, \dots, |P|-1\}$ for each label. Use state squares with lower-left corners

$$o_p = (1/8 + 2si(p), 1/2) \quad (p \neq B_H), \quad o_{B_H} = (3/4, 1/2),$$

and side s . Nonhalting squares lie in $0 < x_1 < 3/8$; the halt square lies in $[3/4, 7/8]$. Use the first planar routing choice above and the sine lift. Load the rational initial code from $(1/2, 1/4)$ along its straight segment, using a translated square cutoff with plateau half-side $1/32$ and support half-side $1/16$. Both endpoints have chart margins at least $1/8$. Nonhalting loading has first coordinate at most $1/2$, and every later nonhalting rectangle path remains in the left half. Thus the event is exact. The main checkpoint times are

$$t_0 = 1, \quad t_{n+1} = t_n + 2(2 + n - h_{n+1}) + 4.$$

For the previous-tag plane use Lemma 10.4 and odd digits in base $2m+1$. Let $r = c(\square^\infty)$ and place the input rectangle at

$$[1/8 - r/16, 1/8 - r/16 + 1/16] \times [1/4 - r/8, 1/4 - r/8 + 1/8].$$

Thus the two blank stacks have code exactly $(1/8, 1/4)$. Halting B -labels have disjoint first intervals inside $(2/3, 5/6)$; all other noninput labels have them inside $(1/4, 1/2)$, with common second interval $[1/4, 1/2]$. To place n labels in (a, b) , take interval $[a + (2j-1)d, a + 2jd]$, $j = 1, \dots, n$, $d = (b-a)/(2n+1)$. Use the logarithmic planar shrink and expansion of the second routing choice and the sine lift. The loader is already a disjoint prefix rule, so period one begins at zero. The first main time is $t_0 = 1$, and the next is $t_{j+1} = t_j + 2|L_j| + 4$. Every rule whose source and target are both nonhalting, including the loader when its target is nonhalting, has both rectangles in the left half and stays there throughout its pulse. Halting main rectangles lie inside the observation arc.

For the shifted-start plane, prepend a fresh nonhalting state q_s which writes back the scanned symbol and stays while entering the original initial state. Apply Lemma 10.8 to this enlarged table. This fresh state is different from the halt state even if the original machine halts initially. Compute the rational initial normalized stack coordinates (ξ_0, η_0) , using blank combined-symbol tails and a history frontier at 2. If L is the number of auxiliary labels, set $\ell = 1/(16(L+1))$. Put the halt square at lower-left corner $(1/8, 1/2)$, the fresh-start square at

$$(3/4, 3/4) - \ell(\xi_0, \eta_0),$$

and the other squares at $(1/2 + (2k+1)\ell, 1/2)$, $k = 0, \dots, L-3$, all of side ℓ . These are disjoint; the initial code is exactly $(3/4, 3/4)$, independently of the input. The ordinary nonhalting squares lie in $1/2 < x_1 < 5/8$, the start square near $(3/4, 3/4)$, and the halt square lies inside $0 < x_1 < 1/3$. Use the third routing choice, with the right half as the safe region and tiny half-side at most $1/100$, then the compact vertical lift about $z = 1/2$. Nonhalting paths have first coordinate greater than $1/3$ throughout. The repeated field starts at zero and is periodic from zero, with no separate loader. The extra initial machine transition is simulated in finitely many periods, so initial halting of the given machine is still detected.

For the separator sine plane use Lemma 10.3 with initial stacks $\#\square^\infty, w\square^\infty$. The nonhalting state rectangles, indexed $j = 1, \dots, n$, are

$$[(3j-2)/(6n), (3j-1)/(6n)] \times [1/4, 3/4],$$

and the halting rectangle is $[2/3, 5/6] \times [1/4, 3/4]$. Use odd digits in base $2m+1$, the moving-center/moving-cutoff fourth routing choice, and the sine lift. A spatially constant planar pulse takes $(1/4, 1/2)$ along the straight segment to the rational initial code during $[0, 1]$. Both endpoints lie in the left half when the initial state is nonhalting, so this loading is safe. Thereafter nonhalting rectangles and their paths remain in the left half; a halt code enters $(1/2, 1)$. The main times are $1 + \sum_{j=1}^k (2|L_j| + 3)$.

For the two-cell-frontier plane use Lemma 10.9, odd digits in base $2m+2$, and its complete three-cell branch list. If N is the number of nonterminal auxiliary labels, set $\ell = 1/(16(N+1))$, put their square origins at $(1/4 + 2i\ell, 1/4)$, $i = 0, \dots, N-1$, and the terminal origin at $(3/4, 1/4)$, all with side ℓ . All source squares belong to nonterminal labels and lie in the left half; the halt square lies strictly in the observation arc. Apply the rational-parabola fifth routing choice and the sine lift. A compact planar loading pulse takes $(1/8, 1/8)$ to the initial code along its segment, with plateau half-side $1/64$ and support half-side $1/32$. The chart margins make this legal, including when initially halted. Nonhalting loading and all nonhalting branches remain in the left half. The main checkpoint recurrence is $t_0 = 1$, $t_{n+1} = t_n + 2(2 + n - h_{n+1}) + 2$.

Finally, for the zero-blank plane use Lemma 10.3, initial stacks $\#\square^\infty, w\square^\infty$, and even digits in base $2m+1$ with blank digit zero. The N nonhalting state squares have centers $(j/(2(N+1)), 1/2)$, $j = 1, \dots, N$, and half-side $1/(8(N+1))$. The halting square has center $(13/16, 1/2)$ and half-side $1/32$. Apply the fixed-cutoff sixth planar routing choice, then the compact vertical lift about $z = 1/2$. The initial normalized code is rational even when on a prefix boundary. A spatially constant planar translation loads it from $(1/4, 1/2)$; that segment stays in the left half in a nonhalting run. All subsequent nonhalting paths stay there as well, while the halt square lies strictly inside $(3/4, 7/8)$. The checkpoint count is again $2|L_j| + 3$ per transition after loading.

For all rows, induction on individual instruction periods now gives the exact encoded auxiliary configuration at each integer boundary through the first halt, or indefinitely in a nonhalting run. The processor lemmas group those samples into the stated machine checkpoints, each after a finite positive number of instructions. In an infinite run the times tend to infinity, so the all-time exclusions

just proved cover every time. After reaching a halted code no further symbolic rule is needed; the smooth fluid motion continues globally. The state rectangles and separated digit intervals recover every finite prefix of both words, hence the represented configurations and their recorded head displacements.

All geometry and initial codes are obtained by finite symbol manipulation and rational arithmetic. Smooth switches, logarithms of positive rational scales, finite differentiations, and bounded-index periodization give effective mixed-derivative evaluation as in Section 2. At no point does constructing or evaluating the force require knowing the successive rules selected by the marked point. The finite pulse acts on every branch domain during every period. The residual realization lemma proves the claimed Navier–Stokes solution and comparison uniqueness. Compact support or spatial periodicity and the repeated finite pulse give uniform derivative and energy bounds.

For the solenoidal alternative in the separator sine-plane row, apply Proposition 2.4 to its mean-zero residual force. If $\Delta r = \operatorname{div} f$ and $\int r = 0$, replace f by $f - \nabla r$ and the pressure by $-r$. In Fourier coordinates,

$$r_k(t) = -\frac{\widehat{\operatorname{div} f}(t, k)}{4\pi^2|k|^2}, \quad k \neq 0, \quad r_0 = 0.$$

Integration by parts gives arbitrarily high inverse-power coefficient bounds for every requested time derivative. The shell $\|k\|_\infty = l$ contains $24l^2 + 2$ modes, so sufficiently many spatial derivatives provide effective summable tails for every derivative of r . The projection preserves period one after loading, mean zero, and bounded derivatives, and leaves the velocity and its event unchanged. $\square$

12.1 Logarithmic clocks for the compact box constructions

The periodic forces above and the following decaying forces are separate choices. A reparametrization changes speed, while preserving every point of the material trajectory.

Proposition 12.1 (Logarithmic alternatives). *For each of the left-word-box, right-word-box, and tagged-storage constructions, retain its fixed initial label and observation set. There is also an effective smooth force of fixed compact spatial support in $L^2([0, \infty) \times \mathbb{R}^3)$ with exactly the same halting event and zero initial velocity. It and all its derivatives are bounded. The velocity’s spatial derivatives are $O((1+t)^{-1})$, and their first time derivatives are $O((1+t)^{-2})$, uniformly in space. The force itself is $O((1+t)^{-1})$ uniformly in space. The viscosity in this statement is the same fixed physical ν as before the time change.*

Proof. Write $U(s, x)$ for the chosen construction, including its loading interval. It is periodic only after that interval, but all its derivatives are uniformly bounded on the whole half-line. Put

$$s(t) = \log(1+t), \quad a(t) = (1+t)^{-1}, \quad \widehat{U}(t, x) = a(t)U(s(t), x).$$

It is smooth and effective and has the same spatial support. Since $U(0) = 0$, its initial velocity is zero. For every spatial multi-index α ,

$$\partial_x^\alpha \widehat{U} = a(\partial_x^\alpha U) \circ s, \quad \partial_t \partial_x^\alpha \widehat{U} = a'(\partial_x^\alpha U) \circ s + a^2(\partial_s \partial_x^\alpha U) \circ s.$$

The uniform derivative bounds for the original finite pulse give the stated spatial and first-time estimates. Repeated chain and product rules express every higher mixed derivative as a finite sum of bounded derivatives of U multiplied by derivatives of a and s , so all are bounded as well. The new residual force is exactly

$$\mathcal{F}_\nu[\widehat{U}] = a'U \circ s + a^2(\partial_s U + (U \cdot \nabla)U) \circ s - \nu a(\Delta U) \circ s.$$

Its pointwise magnitude is at most $C(1+t)^{-1}$. Fixed compact support gives a finite space-time square integral, since $\int_0^\infty (1+t)^{-2} dt < \infty$. The same realization and uniqueness argument applies. If $X(s)$ was the original material path, then $X(s(t))$ solves the new particle ODE by the chain rule. The clock is increasing and maps $[0, \infty)$ onto itself, so the two paths have identical images and exactly the same observation event. A checkpoint at original time s_j is reached at the finite new time $e^{s_j} - 1$; no finite upper bound on the halting time is required to prescribe this clock. $\square$

13 Two compact realizations of the reciprocal recorder

The preceding area-changing constructions use the third coordinate to compensate a planar determinant. For the ordinary one-head recorder, the planar maps already have determinant one. The third coordinate can then serve only to separate instructions during transport. We give two compact Euclidean realizations: one uses four stationary fields in succession, and the other follows entire moving boxes in three phases. Their observer estimates will follow directly from the horizontal motion, without a vertical restriction on the detector.

Use the frontier recorder of Lemma 10.8 throughout this section. Its frontier starts at absolute cell two; every other history cell is blank and every return mark is zero. Its fixed incoming moves and distinguishing written symbols will separate the target rectangles. At checkpoint k the records occupy $2, \dots, k+1$, the frontier is at $k+2$, and transition $k+1$ takes $2(k+2 - h_{k+1}) + 4$ instructions. These properties hold for the same entire partial table as in that lemma.

Choose the combined alphabet $\mathcal{A}$ of the recorder, of size m . Give its letters the even digits $g_a = 0, 2, \dots, 2m-2$, with the fully blank combined letter first. We will use either $B = 2m$ or $B = 2m+1$. For a tape v whose head is at its relative index zero, set

$$x = \sum_{j \geq 1} g_{v_{-j}} B^{-j}, \quad y = \sum_{j \geq 0} g_{v_j} B^{-j-1}, \quad I_a = [g_a/B, (g_a+1)/B].$$

These are the gapped codes of Section 10.7. A state s is placed at $(o_s + x, y, 0)$ for a specified offset o_s . If a rule reads a , writes b , moves by d , and enters s' , its coordinate map relative to the source and target offsets is

d	source rectangle	(x', y')
1	$[0, 1] \times I_a$	$((g_b + x)/B, By - g_a)$
0	$[0, 1] \times I_a$	$(x, y + (g_b - g_a)/B)$
-1	$I_\ell \times I_a$	$(Bx - g_\ell, (g_\ell + (g_b + By - g_a)/B)/B)$

(16)

For the last row there is one branch for each $\ell \in \mathcal{A}$. The target rectangles are respectively $I_b \times [0, 1]$, $[0, 1] \times I_b$, and $[0, 1] \times (g_\ell/B + I_b/B)$. These formulas follow by removing the scanned digit and pushing the written digit. They apply to every point of the displayed closed rectangles. Determinism separates the sources; the fixed incoming displacement and distinguishing written symbol separate the targets. For a left move the target pair (ℓ, b) supplies that distinction. Every linear part is $\text{diag}(\lambda, \lambda^{-1})$.

The code also retains absolute, rather than only head-relative, configuration information at the machine checkpoints. Count the records immediately to the left of the frontier to recover k . The frontier has absolute position $k+2$; its location in the decoded tape therefore recovers the absolute head position. The work track then recovers the original tape. This use of history records is the retained-history version of the reversible-computation idea of Bennett [1]; the positional affine representation follows the generalized-shift viewpoint of Moore [7, 8].

Proposition 13.1 (Two compact periodic processors). *For every deterministic machine and finite input, and every fixed computable viscosity $\nu > 0$, there are effective smooth velocities U and forces f on $\mathbb{R}^3$ with the following properties. Both have all mixed derivatives bounded, support in a compact spatial set independent of time, and period one for $t \geq 1$; $U(0) = 0$. The solution with pressure zero is unique in the class*

$$u \in C([0, T]; H^2) \cap C^1([0, T]; L^2), \quad p \in C([0, T]; H^1), \quad \sup_{[0, T] \times \mathbb{R}^3} (|u| + |\nabla u|) < \infty$$

for every finite T , and its kinetic energy is uniformly bounded. The following are two separate constructions with their stated fixed particle events:

1. *Four successive stationary localized fields, with initial label $(-1, 0, 0)$ and detector $\{x_1 > 0\}$.*
2. *Three successive phases of moving full-box fields, with initial label $(0, 0, 0)$ and detector $\{x_1 < -1\}$.*

In each case the particle enters the detector at some real time if and only if the machine halts. The repeated part of the prescription depends only on the machine; the input appears only in a one-time loading pulse.

Proof. Let R_i, R'_i be the planar source and target rectangles in (16), with centers c_i, c'_i and first scale λ_i . All their side lengths are at most one. If there are no instructions, use the zero repeated field and the same loader described below. Empty pairwise margin tests are omitted for a singleton family.

For the first realization choose $B = 2m$. Put terminal offsets at 4, 7, 10, $\dots$ and all other offsets at $-4, -7, -10, \dots$. Choose private heights $h_i = 4i$. On four successive quarters of a unit period, perform the following autonomous motions, each with flat progress from zero to one:

1. Translate upward by $h_i e_3$, with a cutoff equal to one near $R_i \times [0, h_i]$.
2. At height h_i , use the Hamiltonian vector potential

$$(0, 0, \chi_i \log(\lambda_i)(X_1 - c_{i1})(X_2 - c_{i2})).$$

Its curl on the plateau is $\log(\lambda_i)(X_1 - c_{i1}, -(X_2 - c_{i2}), 0)$.

3. Translate horizontally by $c'_i - c_i$ at height h_i .
4. Translate downward by $h_i e_3$ near $R'_i \times [0, h_i]$.

A translation by w is localized by $\text{curl}(\chi(w \times X)/2)$. In the first and last quarters, choose cutoff collars below one third of the smallest positive horizontal separation in the source and target families, respectively. In the middle quarters, collars of thickness one in height do not meet because the heights differ by four. For the stretch, choose the horizontal plateau to include $c_i + [-1, 1]^2$. For translation, include the bounding rectangle of $[c_i, c'_i]$ enlarged by one in each horizontal direction.

More explicitly, with $\vartheta(s) = \sigma(2s - 1/2)$ and $\vartheta_j(s) = \vartheta(4s - j + 1)$, multiply the sum of the stationary fields for quarter j by $\vartheta'_j(s)$. The tracked rectangle rises, scales by $(\lambda_i^{\vartheta_2}, \lambda_i^{-\vartheta_2})$, translates its center along $[c_i, c'_i]$, and descends. Each intermediate half-width is at most $1/2$, so the entire rectangle remains on the specified plateaus. The positive collars and finite affine norms also give exactness on a positive initial neighborhood. If both endpoint states are nonterminal, both first center coordinates are at most -3 , and the whole tracked path has negative first coordinate.

For the second realization choose $B = 2m + 1$, nonterminal offsets $3, 6, 9, \dots$, and terminal offsets $-3, -6, -9, \dots$. Thicken each rectangle by $[-1, 1]$ in height, and write c_i^0, c_i^1 for the three-dimensional endpoint centers. Each pair within the source family, and within the target family, has a horizontal coordinate gap at least B^{-2} . Use height $6i$ and collar $\epsilon = 1/(4B^2)$. In three successive phases the prescribed affine paths have center and linear part

	$c_i(s)$	$L_i(s)$
lift	$c_i^0 + 6i\Theta e_3$	I
transfer	$(1 - \Theta)c_i^0 + \Theta c_i^1 + 6ie_3$	$\text{diag}(\lambda_i^\Theta, \lambda_i^{-\Theta}, 1)$
lower	$c_i^1 + 6i(1 - \Theta)e_3$	$\text{diag}(\lambda_i, \lambda_i^{-1}, 1)$,

where Θ is ϑ rescaled to the active third. The proof of Lemma 5.1 applies to these exact data: horizontal gaps separate the collars during lift and lowering, while height gaps six separate vertical half-widths $1 + \epsilon$ during transfer. The localized field is explicitly

$$\sum_i \text{curl}_X \left[\chi_i \left\{ \frac{\dot{c}_i \times (X - c_i)}{2} + \frac{(\dot{L}_i L_i^{-1}(X - c_i)) \times (X - c_i)}{3} \right\} \right]. \quad (17)$$

The trace of $\dot{L}_i L_i^{-1}$ is zero. The curl identity in Lemma 3.2 therefore gives the prescribed affine velocity near each moving box, and ODE uniqueness gives its exact full-box motion. A branch with two nonterminal endpoint states has first center coordinate at least three and half-width at most $1/2$; hence its entire path has first coordinate at least $5/2$.

It remains to initialize and to pass from the finite maps to the fluid event. The initial combined tape is blank except at finitely many cells, so its initial code Z_0 is rational. During $[0, 1]$, a localized curl translation with flat progress carries the fixed label to Z_0 along the straight segment. A compact cutoff contains the whole segment with a positive collar. Repeat the chosen instruction pulse on every later unit interval. The time collars make all joins smooth, and only finitely many spatial supports occur. Thus U has the support and derivative bounds in the statement. Set

$$f = \partial_t U + (U \cdot \nabla)U - \nu \Delta U.$$

The residual realization and comparison argument of Section 2 gives the asserted solution and uniqueness; fixed support and bounded U give bounded kinetic energy. The formula can also be returned as $f = f^{(0)} + \nu f^{(1)}$, with $f^{(0)} = U_t + (U \cdot \nabla)U$ and $f^{(1)} = -\Delta U$ effective independently of ν . Evaluation for an arbitrary fixed real viscosity is relative to that parameter.

At each integer $1 + j$ through a halt, or at every such integer in a nonhalting run, the particle is the exact code after j auxiliary instructions. Lemma 10.8 supplies finite positive instruction counts between the original machine checkpoints. In the first realization, a nonhalting loading segment has negative first coordinate, and all subsequent nonterminal branches stay negative; a terminal code has positive first coordinate. In the second, nonhalting loading has nonnegative first coordinate and later nonterminal paths stay above $5/2$ in that coordinate, while a terminal code has first coordinate at most -2 . An initially halted machine is detected at the end of loading. Thus the exclusions apply at every real time, and a halting run enters the detector no later than its terminal checkpoint. No exclusion is needed during a final transition into a terminal state.

All rectangles, cutoffs, scales and finite branch fields are computed from the finite table. Positive rational scales have effective logarithms and exponentials. Smooth switching and derivative bounds permit evaluation at time joins without equality tests. The finite repeated field acts on all branch domains at every period; neither its definition nor its evaluation uses a predicted sequence of machine configurations. $\square$

14 An autonomous processor with a spatial clock

The preceding constructions repeat a time-dependent pulse. We now make that repetition part of an autonomous divergence-free field: a third coordinate runs through the phases of a planar Hamiltonian motion. The observer is restricted to a quiet clock window, so transient transport paths cannot give false detections. An onto scalar time change then starts the fluid from rest and makes its force decay.

Theorem 14.1. *At fixed positive computable viscosity, a deterministic one-tape machine and finite input effectively determine a smooth mean-zero general force on the unit torus such that, for every $k \geq 0$ and spatial multi-index β ,*

$$\|\partial_t^k \partial_x^\beta f(t)\|_\infty \leq C_{k,\beta}(1+t)^{-1}. \quad (18)$$

Its zero-data Navier–Stokes solution is globally smooth and unique in the classical periodic comparison class, with uniformly bounded kinetic energy and mean-zero pressure. For the fixed label and open set

$$y_* = (1/2, 1/2, 1/4), \quad \mathcal{O} = \{(x_1, x_2, z) : 1/2 < x_1 < 1, 0 < z < 1/8\}, \quad (19)$$

its material trajectory enters $\mathcal{O}$ exactly when the machine halts. Samples encode the full machine computation through the first halt, or indefinitely if there is none. The prescription uses only the finite table and input; arbitrary fixed real positive viscosity is allowed as a relative coefficient.

14.1 A recorder with identifiable incoming instructions

Normalize to one halt state q_f and a total nonhalt table. Add a fresh initial state q_I whose instructions preserve the symbol, stay put, and enter the old initial state. No rule enters q_I . This guarantees a positive number of steps before any halt and ensures that the initial control is never visited again. Let Q and A be the normalized state set and work alphabet. Apply the frontier recorder of Lemma 10.8 to this normalized ordinary machine, using exactly its notation:

$$\mathcal{R} = (Q \setminus \{q_f\}) \times A, \quad \Sigma = A \times \{\mathbf{e}, \mathbf{F}, \lambda_r : r \in \mathcal{R}\} \times \{0, 1\}.$$

The main controls are B_q ; the other controls are A_r, R_r, N_q, L_q . At a main checkpoint the work track represents the ordinary tape, all return marks vanish, and a rightward frontier follows the finite history block. One ordinary step writes and moves the work head, marks its new position, records the instruction at the frontier, advances that frontier, and returns to the marked position. Lemma 10.8 proves this cycle and its full-domain inverse with every nonfrontier guard. In particular each target control has a fixed incoming displacement and the written whole symbol determines its incoming instruction. These are the precise hypotheses used for the reciprocal rectangles below.

Initialize at B_{q_I} , head zero, with the input work tape, all marks zero, history frontier $\mathbf{F}$ at cell two and $\mathbf{e}$ elsewhere. At the n th ordinary checkpoint, after primitive count m_n , the work configuration is the normalized machine's n th configuration, records occupy cells $2, \dots, n+1$, and the frontier is at $n+2$. If h_{n+1} is the new work-head position, the same lemma gives

$$m_{n+1} - m_n = 2(n+2 - h_{n+1}) + 4. \quad (20)$$

Thus each ordinary step takes finitely many positive primitive steps. The fresh state q_I occurs only at the initial ordinary checkpoint, and at least one ordinary step precedes any halt. In a nonhalting run no positive primitive sample has main control B_{q_f} or B_{q_I} .

14.2 Reciprocal rectangle maps

Use odd digits and base $K = 2|\Sigma| + 1$ in Lemma 3.1. For a tape relative to its head let $x = E(\sigma_{-1}\sigma_{-2}\cdots)$ and $y = E(\sigma_0\sigma_1\cdots)$, and encode by $o_r + (x, y)$. The two internal initial coordinates $x_{\text{in}}, y_{\text{in}}$ are rational. Choose

$$o_{B_{q_I}} = (-x_{\text{in}}, -y_{\text{in}}), \quad o_{B_{q_f}} = (2, 0), \quad o_r = (-3j, 0) \quad (j = 1, 2, \dots) \quad (21)$$

for the remaining controls. The initial code is zero, and the closed state squares $o_r + [0, 1]^2$ are disjoint.

For an instruction $(r, \eta) \mapsto (r', \beta, d)$ write $h_\sigma(v) = (d(\sigma) + v)/K$. The normalized branches are

$$\begin{aligned} d = 0 : & \quad (x, h_\eta(v)) \mapsto (x, h_\beta(v)), \\ d = 1 : & \quad (x, h_\eta(v)) \mapsto (h_\beta(x), v), \\ d = -1 : & \quad (h_\gamma(w), h_\eta(v)) \mapsto (w, h_\gamma(h_\beta(v))), \quad \gamma \in \Sigma. \end{aligned} \quad (22)$$

All free variables range independently over $[0, 1]$. These are exactly the head-relative list operations. Sources are separated by control, read symbol, and the extra left symbol for a left move. For a common target control the incoming move is fixed and distinct rules have distinct written symbols. Those symbols separate the right first digit for a stay, the left first digit for a right move, and the right pair (γ, β) for a left move. Thus targets are separately separated as well. The linear parts are I , $\text{diag}(1/K, K)$, and $\text{diag}(K, 1/K)$, respectively. Unlike general prefix maps, all preserve planar area. This leaves the third coordinate available for a clock.

14.3 A Hamiltonian extension without planar contraction

The area-preserving requirement rules out the simultaneous contraction used in Lemma 6.1. Instead we separate the rectangle centers, carry the unchanged rectangles to distant parking positions, and deform them there with reciprocal scales.

Lemma 14.2. *Let $U_i, \hat{U}_i$, $1 \leq i \leq J$, $J \geq 1$, be two separately disjoint families of nondegenerate rational closed rectangles in $\mathbb{R}^2$. For each pair prescribe*

$$X \mapsto \hat{c}_i + \text{diag}(\lambda_i, \lambda_i^{-1})(X - c_i), \quad \lambda_i > 0$$

with rational λ_i and the respective centers, and require the compatibility condition

$$\hat{U}_i = \hat{c}_i + \text{diag}(\lambda_i, \lambda_i^{-1})(U_i - c_i).$$

There is an effective smooth compactly supported planar Hamiltonian field $V(\xi, X)$, with time support in $(1/4, 3/4)$, whose unit-time map has this formula on each U_i and a positive neighborhood. The construction gives a rational bound on its spatial support and all controlled rectangle paths.

Proof. Choose a positive rational margin ϵ at most one quarter of a positive coordinate-separating gap for each pair in each family; for empty pair lists take $\epsilon = 1$. The enlarged families stay disjoint. Choose rational D_0 larger than ϵ plus every source or target half-side. Take an integer $G \geq 1$ so that within each family the scaled centers Gc_i and $G\hat{c}_i$ are more than $10D_0$ apart in the sup norm. Choose parking centers e_i more than $10D_0$ apart from one another and from all these scaled centers. A sufficiently distant row with spacing $20D_0$ does this effectively.

Use affine paths

$$X \mapsto C_i(\xi) + \text{diag}(a_i(\xi), a_i(\xi)^{-1})(X - c_i). \quad (23)$$

First translate centers simultaneously $c_i \rightarrow Gc_i$, keeping shapes fixed; any coordinate separation can only increase. Next move one rectangle at a time to e_i . At those parking centers change a_i from 1 to λ_i . Each half-side stays between its endpoint values and hence below $D_0 - \epsilon$. Move the target-shaped rectangles one at a time to $G\hat{c}_i$, then translate centers simultaneously down to $\hat{c}_i$. In this last phase the target separations stay valid down to scale one. During source evacuation, occupied centers are scaled source centers or parking centers. During target delivery, every source has already been evacuated, and occupied centers are parking centers or scaled target centers. Each such mixed family has separation greater than $10D_0$ by the choices above. Thus arbitrary intersections between the original source and target families never place two occupied rectangles at the same location. The obstacle family for each transfer consists of every other currently occupied rectangle, wherever it is in that phase.

For a one-at-a-time transfer, put a closed square of sup radius $4D_0$ around every stationary center. These obstacles are disjoint, and the transfer endpoints are outside them. Follow the straight segment, replacing any portion inside one obstacle by a boundary detour in a fixed orientation. Entry and exit points solve rational linear inequalities, so all waypoints are rational. A tangent contact, a corner contact without entry, or a segment already on the boundary needs no detour: it already has the required clearance. A segment crossing the interior has rational first and last intersection points, found by clipping its rational parameter interval against the four rational side inequalities. A fixed clockwise boundary arc between those points uses only rational corners. The obstacles are separated, so each detour misses every other obstacle. The moving center remains at sup distance at least $4D_0$ from all stationary centers; both enlarged rectangles have radius less than D_0 , proving separation throughout.

Schedule the finitely many submotions in successive equal slots inside $[1/3, 2/3]$, using a flat step for each segment and scale change. This smooths time without changing any path or clearance. Let χ_i equal one on a positive collar of the moving rectangle and be supported in its ϵ enlargement. With $Y = X - C_i$, set

$$\mathcal{H} = \sum_i \chi_i \left(\dot{C}_{i1} Y_2 - \dot{C}_{i2} Y_1 + \frac{\dot{a}_i}{a_i} Y_1 Y_2 \right), \quad V = (\partial_{X_2} \mathcal{H}, -\partial_{X_1} \mathcal{H}). \quad (24)$$

The supports are disjoint. On rectangle i the field is $\dot{C}_i + \text{diag}(\dot{a}_i/a_i, -\dot{a}_i/a_i)(X - C_i)$, so (23) solves its ODE exactly. The same formula holds on a positive initial neighborhood by the plateau and finite affine norm bounds. Smooth ODE uniqueness identifies the motion. A rational support bound follows from the finite endpoint and waypoint list plus $D_0 + 1$. Every operation is effective finite geometry and the fixed flat step, proving the lemma. $\square$

Apply this lemma to (22). With $I_\eta = h_\eta([0, 1])$, the normalized target rectangles for moves $0, 1, -1$ are, respectively, $[0, 1] \times I_\beta$, $I_\beta \times [0, 1]$, and $[0, 1] \times h_\gamma(I_\beta)$. Their side lengths are exactly those of their source rectangles multiplied by $(1, 1)$, (K^{-1}, K) , and (K, K^{-1}) . Adding the respective state offsets verifies the compatibility equation for every branch. The branch list is nonempty because of the fresh initial state. Choose rational $M > 1$ so that its support, controlled paths, and all state squares lie in $(-M, M)^2$. We now have one complete planar transition pulse with its own proof of area preservation; no third-dimensional strain has been used.

14.4 The spatial clock and the observation window

Set $\kappa = 1/(16M)$ and $T(X) = (1/2, 1/2) + \kappa X$. All relevant geometry lies after scaling in $(7/16, 9/16)^2$. Periodically extend

$$\tilde{V}(x_1, x_2, z) = \kappa V(z, T^{-1}(x_1, x_2)).$$

Its known zero collars at the horizontal boundaries and at $z = 0, 1$ make this smooth. It has horizontal divergence zero; each horizontal component has integral zero in (x_1, x_2) since it is a scaled derivative of the compact Hamiltonian.

For an explicit vertical clock choose a flat step θ that is zero on $(-\infty, 1/3]$ and one on $[2/3, \infty)$. Put

$$\psi(s) = (1 - \theta(16(s - 1/8)))(1 - \theta(16(-s - 1/8))), \quad b(x_1, x_2) = \psi(x_1 - 1/2)\psi(x_2 - 1/2),$$

periodize b , and set $w(x_1, x_2) = b(x_1, x_2) - b(x_1 - 1/2, x_2)$. The bump is one on $[3/8, 5/8]^2$ and supported in $(5/16, 11/16)^2$. Its half-period translate has disjoint support, so $w = 1$ on all scaled state squares and controlled paths, and $\int w = 0$. The autonomous field

$$W = (\tilde{V}_1, \tilde{V}_2, w) \tag{25}$$

is divergence free and mean zero. Write Ψ_τ for its global smooth flow on the torus.

From $(T(X), 1/4)$ with X in a source rectangle, the clock increases at speed one and the horizontal coordinates follow its prescribed path with planar phase $\xi = 1/4 + \tau$. After that pulse, the horizontal target stays fixed while the clock passes from $z = 0$ to $z = 1/4$. The entire path has $w = 1$, so this description solves the autonomous ODE. Thus at unit times the map performs one primitive transition and returns the clock to $1/4$. If X_j is the primitive configuration code, induction gives

$$\Psi_j(y_*) = (T(X_j), 1/4) \tag{26}$$

through halt or indefinitely in a nonhalting run. Along these controlled paths, $z = 1/4 + \tau \pmod{1}$. Hence for every positive such j , in the window

$$j - 1/4 < \tau < j - 1/8 \tag{27}$$

the horizontal coordinate is exactly $T(X_j)$ and $0 < z < 1/8$. If the machine halts, its primitive arrival index is positive and its halt square has first offset 2, so this window meets $\mathcal{O}$. If it never halts, the clock test in (19) holds only in these windows. At their positive samples the control is neither B_{q_I} nor B_{q_F} , hence its square has first offset $-3k$, $k \geq 1$, and $T(X_j)_1 < 1/2$. This excludes all false detections, regardless of where intermediate transport paths pass. The samples at m_n encode the normalized work configuration. Discarding its initial stay step recovers the supplied machine's computation. Count the records to recover n , including $n = 0$ at initialization. The frontier has absolute position $n + 2$; subtracting its decoded head-relative index recovers the absolute head position. The work track then gives the absolute tape configuration.

14.5 Starting from rest while covering all internal time

Use the scalar profile

$$A_0(t) = \frac{t}{(1+t)^2} = (1+t)^{-1} - (1+t)^{-2} \tag{28}$$

and prescribe

$$f = A'_0 W + A_0^2 (W \cdot \nabla) W - \nu A_0 \Delta W. \tag{29}$$

Then $(u, p) = (A_0 W, 0)$ solves the forced equation and has $u(0) = 0$. In contrast, $A'_0(0) = 1$ gives $f(0) = W$; the force need not vanish at zero. All three terms have zero spatial mean, using $(W \cdot \nabla) W = \text{div}(W \otimes W)$. Lemma 2.2 gives uniqueness in the periodic comparison class. Since $0 \leq A_0 \leq 1/4$, the kinetic energy is at most $\|W\|_2^2/32$.

For explicit derivative bounds put $B_{r,k} = r(r+1)\cdots(r+k-1)$, with $B_{r,0} = 1$. The k th derivative of $(1+t)^{-r}$ has absolute value $B_{r,k}(1+t)^{-r-k}$. Writing

$$a_k = B_{1,k} + B_{2,k}, \quad b_k = B_{2,k} + 2B_{3,k}, \quad c_k = B_{2,k} + 2B_{3,k} + B_{4,k},$$

and letting $M_{0,\beta}, M_{1,\beta}, M_{2,\beta}$ be the sup norms of the spatial β derivatives of $W, (W \cdot \nabla)W, \Delta W$, respectively, gives

$$\|\partial_t^k \partial_x^\beta f(t)\|_\infty \leq (b_k M_{0,\beta} + c_k M_{1,\beta})(1+t)^{-2-k} + \nu a_k M_{2,\beta}(1+t)^{-1-k}.$$

This proves (18) at every order, including time zero, and supplies the sharper derivativewise decay as well.

The material flow is

$$F(t, y) = \Psi_{S(t)}(y), \quad S(t) = \int_0^t A_0(s) ds = \log(1+t) + (1+t)^{-1} - 1. \quad (30)$$

Differentiation proves the formula. Since $A_0(t) > 0$ for $t > 0$ and $S(t) \rightarrow \infty$, this clock is continuous and strictly increasing from zero onto $[0, \infty)$. Every finite autonomous event and every sample $S^{-1}(j)$ therefore occur at finite physical time. The established window equivalence proves the theorem.

All finite tables, initial geometric tails, rectangle offsets, clearances, waypoints, and scaling constants are effective. The Hamiltonian cutoff uses only rescaled flat steps and positive rational margins, and its periodization has known zero collars. The remaining force operations are finite differentiation and the rational profile (28). The input affects the initial-state translation in (21); no list of executed states is used. At every finite time the material map is a smooth diffeomorphism, with $u = \partial_t F \circ F^{-1}$ and $\partial_t^2 F = (\nu \Delta u - \nabla p + f) \circ F$, as follows by differentiating its trajectory equation.

References

- [1] Charles H. Bennett. Logical reversibility of computation. *IBM Journal of Research and Development*, 17(6):525–532, 1973.
- [2] Robert Cardona, Eva Miranda, and Daniel Peralta-Salas. Computability and Beltrami fields in Euclidean space. *Journal de Mathématiques Pures et Appliquées*, 169:50–81, 2023.
- [3] Robert Cardona, Eva Miranda, Daniel Peralta-Salas, and Francisco Presas. Constructing Turing complete Euler flows in dimension 3. *Proceedings of the National Academy of Sciences*, 118(19):e2026818118, 2021.
- [4] Søren Dyhr, Ángel González-Prieto, Eva Miranda, and Daniel Peralta-Salas. Turing complete Navier–Stokes steady states via cosymplectic geometry. *PNAS Nexus*, 5(5):pgag131, 2026.
- [5] Rolf Landauer. Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development*, 5(3):183–191, 1961.
- [6] Jean Leray. Sur le mouvement d’un liquide visqueux emplissant l’espace. *Acta Mathematica*, 63:193–248, 1934.
- [7] Cristopher Moore. Unpredictability and undecidability in dynamical systems. *Physical Review Letters*, 64(20):2354–2357, 1990.

- [8] Cristopher Moore. Generalized shifts: unpredictability and undecidability in dynamical systems. *Nonlinearity*, 4(2):199–230, 1991.
- [9] OpenAI. Finite Instructions and Solenoidal Shear Flows. OpenAI Math Release preprint [OAI:Finite-Instructions-and-Solenoidal-Shear-Flows-September-27-2026](#), 2026.
- [10] A. M. Turing. On computable numbers, with an application to the Entscheidungsproblem. *Proceedings of the London Mathematical Society*, 42:230–265, 1936–1937.